

Common Venture Capital Ownership and Startup Innovation

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Abstract

Common venture capital (VC) ownership, in which a startup shares a VC investor with rival startups, is increasingly prevalent. Yet whether such ownership promotes or inhibits startup innovation remains unclear. Common VC ownership can affect startup innovation in opposing ways: by expanding startups' access to useful knowledge and resources, it can facilitate innovation, but by increasing concerns that proprietary information may reach rival portfolio startups, it can also dampen innovation. Using the staggered adoption of Corporate Opportunity Waivers as an instrument for common VC ownership, I find that common VC ownership increases startup innovation on average, as reflected in both patenting and subsequent patent citations. Common VC ownership also increases technological similarity and patent cross-citations, consistent with stronger technological connections among startups. These innovation gains vary systematically with the startup's exposure to knowledge leakage. They are smaller when the common VC investor is geographically close to a rival startup it also backs and larger in industries with stronger intellectual property protection. The gains also vary with the startup's market position. The positive effect on innovation is largest when technological similarity is high and product market similarity is low, whereas the effect becomes negative when technological similarity is low and product market similarity is high. These findings suggest that whether common VC ownership benefits or harms startup innovation depends on both exposure to knowledge leakage and the technological and competitive relationships among portfolio startups.

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1 Introduction

Common venture capital (VC) ownership, in which a startup shares a VC investor with rival startups, has become increasingly prevalent over the past two decades (Eldar and Grennan, 2021, 2024). Prior to their initial public offerings (IPOs), for example, Uber and Lyft shared at least five common VC investors. Prior research suggests that common VC ownership can meaningfully shape startups’ development choices and strategic behavior (Li et al., 2023; Cox Pahnke et al., 2015). However, its overall effect on startup innovation, and the conditions under which it benefits or harms innovation, remain unclear.

Founders themselves are divided on this question. Nick Evans, co-founder of Avocado, a restaurant point-of-sale startup, captures the optimistic view: *“I want investors that deeply understand my business and industry. How the hell would they know anything useful if they aren’t working with similar companies?”*¹ Others view common VC ownership as a potential information hazard. As Chris Neumann, General Partner at Panache Ventures, puts it, *“a VC with a board seat in your competitor will (almost certainly) share any information you give them with your competitor.”*²

These competing views reflect two opposing forces. On the one hand, a common VC investor can be a source of *knowledge facilitation*. From the startup’s perspective, the investor may help transfer technological know-how across portfolio firms facing similar challenges (Polanyi, 1966; González-Uribe, 2020), provide more targeted guidance drawn from experience with related ventures (Gorman and Sahlman, 1989; Kaplan and Strömberg, 2003), and channel resources more effectively through specialization across a focused set of investments (Lindsey, 2008). On the other hand, the same structure can trigger *knowledge dampening*. Because startups are implicitly connected through the common VC investor, a startup may worry that proprietary information disclosed to the investor will reach a rival also backed by that investor (Cox Pahnke et al., 2015). Anticipating this risk, the startup may restrict disclosure and divert resources and entrepreneurial

¹<https://techcrunch.com/2024/11/22/y-combinator-often-backs-startups-that-duplicate-other-vc-companies-data-shows-its-not-just-ai-code-editors/>

²<https://chrisneumann.com/archives/dont-talk-to-your-competitors-investors>.

attention toward protecting information rather than producing innovation ([Anton and Yao, 1994](#); [Horstmann et al., 1985](#); [Ziedonis, 2004](#); [Somaya, 2003](#); [Gans et al., 2026](#)).

In this paper, I examine how common VC ownership affects the rate and direction of startup innovation and the conditions under which its benefits are more likely to outweigh its costs. Ex ante, it is unclear whether knowledge facilitation or knowledge dampening will dominate. I argue that which force dominates varies systematically with two sets of boundary conditions. The first is the startup's *exposure to knowledge leakage*. I characterize this exposure using geographic proximity between the common VC investor and rival startups, which affects the likelihood that information disclosed by the focal startup reaches a rival, and intellectual property (IP) protection, which affects the expected harm if such leakage occurs by shaping the startup's ability to appropriate returns from innovation.

The second set of boundary conditions concerns the startup's *market position* relative to rival startups backed by the common VC investor. I focus on two dimensions: *technological similarity* and *product market similarity*. Technological similarity captures overlap in the knowledge bases, capabilities, and technical approaches that startups use to create their offerings, whereas product market similarity captures the extent to which those offerings are substitutes ([Bloom et al., 2013](#); [Hoberg and Phillips, 2016](#)). Greater technological similarity increases both the relevance of knowledge available through the common VC investor and the investor's incentive to facilitate its use, even though it may also heighten concerns about knowledge leakage. Greater product market similarity, by contrast, increases the competitive harm from leakage and weakens the investor's incentive to mitigate those concerns. Technological similarity should therefore shift the balance toward knowledge facilitation, whereas product market similarity should shift it toward knowledge dampening.

Empirically examining the effect of common VC ownership on startup innovation and the conditions that shape this effect requires addressing three challenges. The first challenge is measuring common VC ownership. Prior studies often capture common VC ownership at the extensive margin, using an indicator for whether a startup shares a VC investor with another startup in the same

industry (Eldar and Grennan, 2024; Cox Pahnke et al., 2015). This binary classification treats all same-industry startups as equally relevant rivals, even though some are much more closely related to the focal startup than others. I therefore construct a continuous measure of *CommonVC* that captures variation in the economic relevance of rivals connected to a startup through common VC investors. The measure combines patent information from PatentsView with business descriptions from PitchBook and VentureXpert to distinguish economically closer shared-VC relationships from more distant ones (Jaffe, 1986; Bloom et al., 2013; Hoberg and Phillips, 2016). This approach captures an important intensive margin of common VC ownership that is missed by a binary industry-based measure.³

The second challenge is measuring a startup’s market position along the two dimensions of technological similarity and product market similarity. Unlike public firms, startups generally do not provide standardized financial disclosures or detailed information about their products and technological activities, making their technological and product market positions difficult to observe. I address this challenge by measuring technological similarity using startup patent data (Ewens and Marx, 2023) and product market similarity using United States Patent and Trademark Office (USPTO) trademark data. Technological similarity is based on overlap in the technological fields represented in startups’ patent portfolios (Bloom et al., 2013), whereas product market similarity is based on overlap in the goods and services associated with their trademarks (Gao and Hitt, 2012; Hsu et al., 2022a). Together, these measures allow me to separately characterize startups’ positions in technological and product market space and examine how these two dimensions shape the effect of common VC ownership on innovation.

The third challenge is identification because VC investors do not invest in startups randomly. VC investors with superior selection capabilities may invest in startups with greater innovation potential, while more innovative startups may themselves attract greater VC interest (Sørensen, 2007). These selection effects can generate a positive association between common VC ownership and startup innovation even in the absence of a causal effect. I address this challenge by exploit-

³As a robustness check, Appendix Table II replaces *CommonVC* with an indicator for whether the startup shares at least one VC investor with another startup in the same industry. The main results remain unchanged.

ing the staggered adoption of Corporate Opportunity Waivers (COWs) across nine states in the United States (U.S.) between 2000 and 2016 as an instrument for common VC ownership. Before COWs, the Corporate Opportunity Doctrine (COD) exposed VC investors to legal and reputational risks when they invested in rival startups, discouraging common VC ownership ([Atanasov et al., 2012](#); [Eldar and Grennan, 2024](#)). COWs allowed corporations to waive the associated corporate-opportunity liability, reducing this legal friction to common VC ownership ([Eldar and Grennan, 2024](#)). Several features strengthen the identification strategy. COW eligibility is determined by a startup's state of incorporation rather than its headquarters or operating location, making adoption less directly tied to local changes in startup innovation. The statutes also narrowly modify the COD rather than introducing broader changes to corporate governance or innovation policy, and legislative records show no evidence that VC lobbying drove adoption ([Eldar and Grennan, 2024](#)). Together, these features reduce the concern that COW adoption reflects changes in startup innovation or demand for common VC ownership that were already underway.

Using this identification strategy, I find that common VC ownership increases startup innovation on average, as measured by both patent counts and five-year forward patent citations. The increase in forward citations helps mitigate the concern that greater patenting primarily reflects defensive patent accumulation, which could raise patent counts without a corresponding increase in technological value ([Ziedonis, 2004](#)). The results are robust across alternative samples and measures, and a placebo test shows no significant effect of COW adoption among startups unlikely to face a plausible rival, for which the policy should have little effect on common VC ownership.⁴ I further find that common VC ownership increases technological similarity and, among connected startups, patent cross-citations. Together, these patterns provide complementary evidence consistent with knowledge facilitation. Moreover, COW adoption increases director interlocks among connected startups, providing an observable governance channel through which common VC investors can remain directly involved across multiple portfolio firms.

The positive average effect, however, does not imply that the benefits of common VC ownership

⁴These robustness checks are reported in Appendix Tables [A2–E1](#).

outweigh its costs for all startups. I first examine heterogeneity in a startup's exposure to knowledge leakage. The innovation gains from common VC ownership are weaker for startups whose common VC investor is co-located with a rival startup it also backs. Such proximity increases opportunities for information disclosed to the investor to reach the rival, heightening the startup's exposure to leakage (Jaffe et al., 1993; Audretsch and Stephan, 1996; Bernstein et al., 2016). By contrast, the gains from common VC ownership are stronger for startups in industries with stronger IP protection, where stronger appropriability reduces the expected harm if information reaches a rival (Graham et al., 2009; Gans and Stern, 2003; Anton and Yao, 2004; Hall and Harhoff, 2012).

Beyond exposure to knowledge leakage, the effect of common VC ownership also varies with the startup's market position relative to rivals. Consistent with the argument that technological similarity shifts the balance toward knowledge facilitation whereas product market similarity shifts it toward knowledge dampening, the innovation gains are largest when technological similarity is high and product market similarity is low. When both dimensions are high, common VC ownership continues to increase innovation, but the gains are smaller. When both dimensions are low, the effect remains positive but is substantially smaller in magnitude. Finally, when technological similarity is low and product market similarity is high, the effect becomes negative for citation-based measures of innovation.

This paper makes three contributions. First, I contribute to the emerging literature on common VC ownership by explaining when common VC ownership is more likely to benefit or harm startup innovation. Prior work documents both potential benefits and costs of common VC investors: startups that share VC investors with industry peers grow faster and exit more successfully (Eldar and Grennan, 2024), common VC investors can consolidate overlapping development efforts (Li et al., 2023), and connections through a common VC investor can expose startups to information leakage and reduce new product introduction (Cox Pahnke et al., 2015; Leccese, 2023). What remains less understood is why these benefits dominate in some settings while the costs dominate in others. I bring these competing forces together in a framework of knowledge facilitation and knowledge dampening and identify two sets of boundary conditions: a startup's exposure to knowledge leakage

and its market position. This framework contributes to the literature by moving beyond whether common VC ownership is beneficial or harmful on average and explaining the conditions under which each effect is more likely to dominate.

Second, I contribute to the broader literature on common ownership and innovation by examining a setting in which common owners exercise substantial influence over the firms they own. Existing evidence from public firms is mixed: some studies find that common ownership increases research and development (R&D) and innovation ([Antón et al., 2023](#); [He and Huang, 2017](#); [Geng et al., 2015](#); [Antón et al., 2025](#)), whereas others emphasize reduced R&D incentives and anticompetitive effects ([Bindal, 2022](#); [Geng et al., 2022](#); [Kostovetsky and Manconi, 2020](#); [Azar et al., 2018](#)). Institutional investors in public markets typically hold relatively small stakes and exercise limited direct control ([Gerardi et al., 2023](#); [Geng et al., 2022](#)). VC investors, by contrast, hold concentrated stakes, exercise strong control rights, and participate actively in startup governance ([Lerner, 1995](#); [Kaplan and Strömberg, 2003](#)). The venture setting therefore highlights a role of common ownership that is less visible in public markets: a common owner can directly shape the knowledge and resources available to portfolio firms while also creating a channel through which information may reach rivals.

Third, I contribute to research in corporate strategy on interorganizational knowledge exchange by extending attention beyond deliberately formed bilateral relationships. Prior work largely examines alliance partners that choose to collaborate and use contractual or relational safeguards to manage the tension between knowledge exchange and knowledge hazards ([Dyer and Singh, 1998](#); [Powell et al., 1996](#); [Asgari et al., 2026](#)). Under common VC ownership, by contrast, portfolio startups need not select one another as partners or enter into direct agreements with one another. Instead, they are connected through a common investor that can shape the knowledge and resources available across portfolio firms. I show that such investor-mediated connections can generate both benefits from access to useful knowledge and risks that proprietary information reaches rival portfolio startups. This extends the literature by identifying common owners as an important third-party governance mechanism through which interorganizational knowledge exchange can be facilitated

and constrained even in the absence of a direct collaborative relationship between firms.

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework. Section 3 provides background on COWs. Section 4 describes the data and measurement. Section 5 outlines the empirical strategy. Section 6 presents the results. Section 7 concludes.

2 Theory

Common VC ownership creates two opposing forces for startup innovation. By expanding the knowledge and resources available to the focal startup, it can facilitate innovation. At the same time, by creating a route through which proprietary information may reach rival portfolio startups, it can dampen innovation. I refer to these forces as *knowledge facilitation* and *knowledge dampening*. Because both forces can operate simultaneously, the effect of common VC ownership on startup innovation is ambiguous ex ante. I first develop these two forces and then examine the conditions that shift the balance between them.

2.1 Knowledge Facilitation and Knowledge Dampening

Knowledge facilitation. Common VC ownership can increase startup innovation by expanding the knowledge and resources available to the focal startup. VC investors play an active governance role through board participation, strategic advice, and repeated interaction with portfolio firms (Gorman and Sahlman, 1989; Lerner, 1995; Kaplan and Strömberg, 2003). By investing across multiple startups, a common VC investor accumulates domain expertise and develops a portfolio-wide view of technical problems and potential solutions. This position allows the investor to provide informed guidance and resources (Sørensen, 2007; Gompers et al., 2009) and connect portfolio startups to useful expertise and other non-financial resources (Hsu, 2004; Hellmann and Puri, 2002; Bernstein et al., 2016; Ozmel et al., 2013).

The common VC investor's position can be particularly valuable for knowledge that is difficult to codify (Polanyi, 1966). Technical experience, problem-solving knowledge, and lessons from prior experimentation may be difficult to transfer fully through patents, publications, or formal

agreements ([Von Hippel, 1994](#); [Szulanski, 1996](#)). Repeated interaction with portfolio startups allows the common VC investor to accumulate and transmit such knowledge through its governance role.

Knowledge dampening. The same connection through a common VC investor can also create a risk of information leakage. A common VC investor receives sensitive information from multiple portfolio startups, creating the possibility that information disclosed by the focal startup reaches a rival backed by the same investor. Prior research shows that investor ties can expose startups to information leakage and reduce new product introduction when connected firms compete ([Cox Pahnke et al., 2015](#); [Fulghieri and Sevilir, 2009](#)).

The information at risk can include both technological and commercial knowledge, such as experimental results, development priorities, product design, customer demand, and commercialization plans. Anticipating that such information may reach a rival, the focal startup may become more cautious about what it reveals to the common VC investor. It may devote resources to defensive patenting and other efforts to protect its proprietary position ([Ziedonis, 2004](#); [Somaya, 2003](#)), or divert managerial attention from experimentation and technology development toward protecting sensitive information ([Gans and Stern, 2003](#); [Gans et al., 2019, 2026](#)). These responses can reduce the resources and attention devoted to innovation.

Common VC ownership therefore creates an inherent tension. The common VC investor can expand the knowledge and resources available to the focal startup, while the possibility that proprietary information reaches a rival can induce defensive responses that dampen innovation. Whether common VC ownership ultimately increases or decreases startup innovation depends on the relative strength of these two forces.

2.2 Boundary Conditions

The balance between knowledge facilitation and knowledge dampening should vary systematically across startups. I consider two sets of boundary conditions. The first concerns the startup's exposure to knowledge leakage, which depends on both the likelihood that disclosed information reaches

a rival and the expected harm if leakage occurs. The second concerns the startup's market position relative to rival startups, which shapes both the value of knowledge available through the common VC investor and the competitive harm associated with knowledge leakage.

2.2.1 Exposure to Knowledge Leakage

I characterize the startup's exposure to knowledge leakage using two conditions: geographic proximity between the common VC investor and a rival startup it also backs, which affects the likelihood that information reaches the rival, and the strength of IP protection, which affects the expected harm if leakage occurs.

Geographic proximity increases the likelihood that information disclosed to the common VC investor reaches a rival startup. Knowledge flows decline with distance, particularly for tacit knowledge that is difficult to codify and transmit across organizational boundaries ([Jaffe et al., 1993](#); [Audretsch and Stephan, 1996](#); [Storper, 1995](#); [Thompson and Fox-Kean, 2005](#)). When the common VC investor is geographically close to a rival startup it also backs, more frequent interaction and overlapping local networks create greater opportunities for knowledge to flow between the investor and the rival ([Bernstein et al., 2016](#)), increasing the focal startup's exposure to knowledge leakage. Greater exposure strengthens knowledge dampening by giving the focal startup stronger incentives to protect sensitive information and divert managerial attention and resources away from experimentation and innovation. The focal startup should therefore experience smaller innovation gains from common VC ownership when the common VC investor is geographically close to a rival startup it also backs.

IP protection affects the expected harm if information reaches a rival. Stronger IP protection makes imitation more difficult and increases the startup's ability to appropriate returns from its innovations ([Teece, 1986](#); [Graham et al., 2009](#)). Stronger protection therefore reduces the potential harm associated with knowledge leakage and weakens the startup's incentives to guard sensitive knowledge. The startup can engage more openly with the common VC investor, reducing knowledge dampening while allowing it to benefit more from knowledge facilitation ([Gans and Stern, 2003](#)). Common VC ownership should therefore generate larger innovation gains when IP protec-

tion is stronger.

2.2.2 Market Position

The second set of boundary conditions concerns the focal startup's market position relative to rival startups backed by the common VC investor. I distinguish two dimensions: *technological similarity*, which reflects overlap in the startups' knowledge bases, capabilities, and technical approaches, and *product market similarity*, which reflects the extent to which their commercialized offerings are substitutes (Jaffe, 1986; Bloom et al., 2013; Hoberg and Phillips, 2016). These dimensions capture distinct aspects of rivalry and can vary independently, with different implications for the balance between knowledge facilitation and knowledge dampening.

Technological similarity should shift the balance toward knowledge facilitation by increasing both the relevance of knowledge available through the common VC investor and the investor's incentive to facilitate its use. Knowledge is more valuable when it applies to the focal startup's technical problems, and a startup can more readily recognize, absorb, and apply knowledge that builds on capabilities and prior understanding similar to its own (Cohen and Levinthal, 1990). Moreover, because technical knowledge is nonrival in use, its application by one startup does not preclude its use by another (Arrow, 1962; Romer, 1990). When portfolio startups face related technical problems, the common VC investor can therefore create value across multiple investments by providing relevant expertise or facilitating the use of knowledge originating elsewhere in the portfolio (Lindsey, 2008; González-Uribe, 2020). Technological similarity therefore both increases the value of knowledge facilitation and strengthens the common VC investor's incentive to support it, shifting the effect of common VC ownership toward knowledge facilitation.⁵

Product market similarity, by contrast, should shift the balance toward knowledge dampening. When two portfolio startups offer close substitutes, they compete more directly for the same cus-

⁵Technological similarity may also heighten knowledge leakage concerns because a technologically related rival is better positioned to understand and apply leaked technical information (Cohen and Levinthal, 1990; Jaffe, 1986; Bloom et al., 2013), particularly when technologically similar startups compete for overlapping exit opportunities (Bena and Li, 2014). However, this countervailing force should be relatively limited because knowledge exchange among technologically related startups can increase the value of multiple portfolio investments, giving the common VC investor stronger incentives to preserve startups' willingness to disclose information and mitigate leakage concerns (Fulghieri and Sevilir, 2009; Lindsey, 2008; González-Uribe, 2020; Antón et al., 2025).

tomers and product market demand, so gains to one startup are more likely to come at the expense of the other (Bloom et al., 2013; Antón et al., 2023). Information that reaches a rival can therefore help it improve a competing offering and capture customers or sales from the focal startup, increasing the potential harm from knowledge leakage (Bloom et al., 2013; Cox Pahnke et al., 2015). The focal startup consequently has stronger incentives to protect sensitive information and limit what it reveals to the common VC investor. At the same time, because gains among close product market rivals are more likely to redistribute value within the portfolio rather than increase the value of multiple investments, the common VC investor has less to gain from mitigating these concerns (Antón et al., 2025). Product market similarity therefore both increases the competitive harm from knowledge leakage and weakens the investor’s incentive to mitigate it, shifting the effect of common VC ownership toward knowledge dampening.

Taken together, technological similarity tends to shift the balance toward knowledge facilitation, whereas product market similarity tends to shift it toward knowledge dampening. Crossing these dimensions yields four market positions: the *knowledge corridor*, *coordinated rivals*, *faint tailwind*, and *contested market*. Table 1 summarizes the resulting predictions.

The *knowledge corridor* combines high technological similarity with low product market similarity. High technological similarity increases the relevance of knowledge available through the common VC investor and strengthens the investor’s incentive to facilitate its use, while low product market similarity limits the competitive harm from knowledge leakage. Commonwealth Fusion Systems (CFS) and SpaceX provide a useful example. Both are backed by Brightwing Capital and face overlapping engineering challenges involving cryogenic systems and high-performance hardware, yet they commercialize their technologies in distinct markets. CFS develops fusion power systems, whereas SpaceX provides launch and space transportation services. Knowledge relevant to one firm can therefore have technical value for the other without creating equally strong product market conflict. Knowledge facilitation should therefore outweigh knowledge dampening, producing the strongest positive effect of common VC ownership on startup innovation.

When both technological similarity and product market similarity are high, *coordinated rivals*

face strong knowledge facilitation and strong knowledge dampening. This tension is apparent in the relationship between CFS and Proxima Fusion, both backed by Brevan Howard Macro Venture Fund. Both pursue magnetic-confinement fusion using high-temperature superconducting magnet technologies and seek to commercialize fusion power. Their technological similarity makes knowledge available through the common VC investor particularly relevant, but their product market similarity also raises the competitive harm from knowledge leakage and weakens the investor's incentive to mitigate those concerns. Because the two forces operate strongly in opposite directions, the effect of common VC ownership on startup innovation is ambiguous *ex ante*.

The *faint tailwind* combines low technological similarity with low product market similarity, leaving mainly the general benefits of an engaged VC investor while limiting the additional value of knowledge available through the common VC investor. CFS and Zipline, both backed by Evolution VC Partners, capture this combination. CFS develops magnetic-confinement fusion systems, whereas Zipline develops autonomous drone-delivery systems. Their technologies address substantially different technical problems, and their commercial offerings serve different markets. Knowledge available through their common VC investor is therefore less likely to be directly relevant across the two firms, but information leakage is also less likely to create substantial competitive harm. With both knowledge facilitation and knowledge dampening limited, common VC ownership should have only a small or null effect on startup innovation.

Finally, the *contested market* combines low technological similarity with high product market similarity. CFS and General Fusion, both backed by Evolution VC Partners, show how firms can compete in similar markets while relying on substantially different technical approaches. Both seek to commercialize fusion power, but CFS uses high-temperature superconducting magnets in a tokamak design, whereas General Fusion pursues Magnetized Target Fusion based on mechanical compression and liquid-metal systems. The firms can therefore compete for similar commercial opportunities even though the technical knowledge underlying their approaches differs substantially. Knowledge available through the common VC investor is less likely to be directly useful across the two firms, while product market competition increases the potential harm from knowledge leakage

and weakens the investor’s incentive to mitigate those concerns. Knowledge dampening should consequently outweigh knowledge facilitation, making common VC ownership more likely to reduce startup innovation.

Insert Table 1 about here.

3 Institutional Background

3.1 Legal Frictions Before Corporate Opportunity Waivers (COWs)

For much of the modern VC era, the Corporate Opportunity Doctrine (COD) created legal risks for VC investors that held stakes in competing startups. Under the COD, a fiduciary who became aware of a business opportunity within a corporation’s line of business could be required to present that opportunity to the corporation before pursuing it personally or through another entity ([Rauterberg and Talley, 2017](#)). This doctrine was particularly relevant for VC investors because they commonly hold substantial equity stakes, take board seats, and receive sensitive information from their portfolio companies ([Lerner, 1995](#); [Kaplan and Strömberg, 2003](#)). Investing in another startup pursuing an overlapping opportunity could therefore expose a VC investor to claims of corporate opportunity usurpation, breach of fiduciary duty, or improper use of confidential information ([Atanasov et al., 2012](#)).

These risks were especially salient when portfolio startups competed with one another. In *Ajaxo Inc. v. E*Trade Group Inc.*, for example, a startup alleged that an investor disclosed its proprietary information to another portfolio company, resulting in prolonged litigation ([Atanasov et al., 2012](#)). More generally, a VC investor involved with startups pursuing overlapping opportunities faced the possibility of ex post scrutiny over whether opportunities or information belonging to one portfolio company had been improperly used to benefit another. Even when such claims ultimately failed, litigation could impose substantial legal costs, consume managerial attention, and create conflict between the investor and its portfolio companies.

The potential reputational costs further increased these legal frictions. VC is a relationship-

intensive business in which future deal flow depends on trust among investors, founders, and co-investors ([Hallen et al., 2014](#); [Hsu, 2004](#)). A VC firm involved in disputes over conflicts of interest or proprietary information could therefore bear costs beyond the litigation itself. Survey evidence indicates that both VC investors and founders viewed corporate opportunity liability as sufficiently important to discourage investments in competing portfolio companies ([Atanasov et al., 2012](#); [Eldar and Grennan, 2024](#)). Before the adoption of Corporate Opportunity Waivers, the COD therefore raised the expected cost of common VC ownership and reduced the scope for startups to share investors with rival startups.

3.2 What COWs Changed

Between 2000 and 2016, nine U.S. states enacted statutes permitting COWs, beginning with Delaware in 2000. Table 2 reports the states and adoption dates. A COW allows a corporation to renounce its claim to specified business opportunities that would otherwise fall within the COD, reducing the potential liability associated with pursuing those opportunities through another entity ([Rauterberg and Talley, 2017](#)). COW eligibility is determined by a firm’s state of incorporation, so startups incorporated in adopting states became eligible for these waivers following passage.

Insert Table 2 about here.

Several features of COW adoption support its use as a source of plausibly exogenous variation in common VC ownership. First, [Rauterberg and Talley \(2017\)](#) attribute the legislative changes to efforts to resolve doctrinal uncertainty created by corporate opportunity case law rather than to demands from the VC industry. Consistent with this account, [Eldar and Grennan \(2024\)](#) find no evidence of VC lobbying in the legislative records. Second, the statutes specifically changed corporations’ ability to waive corporate opportunity claims rather than broadly altering startup financing, innovation policy, or other dimensions of corporate governance. Third, survey evidence indicates that waivers became common among VC-backed firms once they were legally permitted ([Atanasov et al., 2012](#); [Eldar and Grennan, 2024](#)). Together, these features suggest that COW passage reduced a legal friction to common VC ownership for reasons largely unrelated to startup innovation.

Post-adoption case law illustrates how these waivers changed the legal environment for competing investments. ABS Capital held a stake and board representation in Alarm.com before subsequently investing in two of Alarm.com’s competitors. Alarm.com sued, arguing that confidential information accumulated by the investor could reach those rivals. The Delaware Court of Chancery dismissed the corporate opportunity claim because Alarm.com’s charter contained a waiver, holding that the waiver permitted the investor to make competing investments.⁶ The case illustrates how COWs lowered the legal barrier to holding stakes in competing portfolio companies.

COW adoption should therefore increase startups’ exposure to common VC ownership. Because common VC ownership is measured continuously, the relevant variation is on the intensive margin: startups can differ in how strongly they are connected to rival startups through common VC investors. COW passage reduces the legal cost of such overlapping investments and thereby shifts startups along this margin rather than simply changing a binary treatment status.

4 Data and Sample

4.1 Data Construction

I construct a panel of U.S. VC-backed startups that links their financing histories, VC investors, patenting activities, trademark registrations, and states of incorporation. The data come from six primary sources: PitchBook, VentureXpert, PatentsView, OpenCorporates, the USPTO Trademark Case Files Dataset, and the startup patent database from [Ewens and Marx \(2023\)](#).

I combine PitchBook and VentureXpert to construct startup financing histories. PitchBook provides the primary source of investment information, while VentureXpert provides broader coverage in the earlier years of the sample, which is important because the staggered adoption of COWs begins in 2000. I merge the two datasets using fuzzy matching on startup names, including legal names, together with headquarters information, and append startups identified in VentureXpert that

⁶*Alarm.com Holdings, Inc. v. ABS Capital Partners, Inc.*, C.A. No. 2017-0583-JTL (Del. Ch. June 15, 2018), aff’d, No. 360, 2018 (Del. Feb. 7, 2019). For a discussion, see <https://corpgov.law.harvard.edu/2021/06/29/alarm-com-and-the-open-questions-regarding-trade-secret-claims-related-to-usurpation-of-corporate-opportunities/>.

do not appear in PitchBook.

I link the financing records to patent and trademark data. I use the startup patent database from [Ewens and Marx \(2023\)](#) to identify patents assigned to private VC-backed startups and link these patents to PatentsView for information on application dates, technology classifications, and citation networks. USPTO trademark records are used to characterize startups' positions in product markets. I restrict the sample to U.S. VC-backed startups with at least one patent, and 41.8% of these startups are matched to at least one product trademark in the USPTO data. Because COW eligibility is determined by a startup's state of incorporation rather than its headquarters location, I obtain incorporation-state information from OpenCorporates and supplement these records using VentureXpert.⁷

I define a startup's entry year as its founding year, or its first investment year when the founding year is unavailable. The exit year is the year of the first IPO or merger or acquisition (M&A) event. If no formal exit is observed, I follow [Li et al. \(2023\)](#) and treat a startup as having exited the VC market after five consecutive years without new financing. Because the empirical design relies on COW eligibility, I restrict the sample to startups with observed states of incorporation.

The final panel spans 1990 to 2021 and contains 17,189 VC-backed startups and 170,809 startup-year observations. Of these, 15,635 startups are identified in PitchBook and an additional 1,554 are added from VentureXpert.

4.2 Measuring Innovation and Patent Cross-Citations

I measure startup innovation using patent counts and subsequent patent citations. $Patent\ Count_{it}$ is the number of patents granted to startup i in year t . $5-Year\ Citation_{it}$ counts citations received by these patents within five years of grant, while $5-Year\ Citation\ (NoSelf)_{it}$ excludes self-citations. Excluding self-citations provides a more conservative measure of external technological impact by focusing on citations made by other firms rather than subsequent patents from the focal startup itself.

⁷I gratefully acknowledge OpenCorporates for providing free academic access through its Public Benefit Project; the data are available at <https://opencorporates.com/use-cases/academia>.

Beyond these innovation outcomes, I construct $Cross\ Cite_{it}$ to measure patent cross-citations between the focal startup and startups connected to it through common VC investors. The measure counts citations in both directions between their patents and captures technological referencing among startups connected through common VC ownership.

4.3 Measuring Technological and Product Market Similarity

The theory characterizes startup market position along two dimensions: technological similarity and product market similarity. Technological similarity captures overlap in the knowledge bases and technical capabilities through which startups create value, whereas product market similarity captures overlap in the offerings through which startups commercialize that value. I construct each dimension first at the startup-pair level and then aggregate it across other active startups to characterize the focal startup’s market position.

4.3.1 Technological Similarity

I measure technological similarity using startups’ patent portfolios. Following [Jaffe \(1986\)](#) and [Bloom et al. \(2013\)](#), I construct a technology vector for each startup based on the distribution of its patents across Cooperative Patent Classification (CPC) subclasses:

$$T_{it} = (T_{it1}, T_{it2}, \dots, T_{it540}),$$

where $T_{it\tau}$ is the share of startup i ’s patents assigned to technology class τ up to year t . Pairwise technological similarity is measured using the cosine similarity between the technology vectors:

$$TEC_{ijt} = \frac{T'_{it}T_{jt}}{\sqrt{T'_{it}T_{it}}\sqrt{T'_{jt}T_{jt}}}.$$

Higher TEC_{ijt} indicates greater overlap in the technological knowledge bases of startups i and j .

I also construct a Mahalanobis measure of pairwise technological similarity, $MALTEC_{ijt}$, following [Bloom et al. \(2013\)](#). Whereas the Jaffe measure relies on direct overlap across CPC sub-

classes, the Mahalanobis measure also captures relationships between technologies that frequently occur together across firms. This feature is particularly useful for startups, whose patent portfolios are often small and sparse, making direct class overlap difficult to observe even when their technologies are related.

I aggregate pairwise technological similarity across other active startups to construct the focal startup's technological similarity:

$$\text{Technological Similarity}_{it} = \sum_{j \neq i} TEC_{ijt}.$$

Higher Technological Similarity indicates that the focal startup's technological knowledge base overlaps more extensively with those of other active startups. I construct *Technological Similarity (Mahalanobis)* analogously by replacing TEC_{ijt} with $MALTEC_{ijt}$.

4.3.2 Product Market Similarity

I measure product market similarity using USPTO trademark records. Trademarks identify the goods and services through which firms commercialize their activities and therefore provide a direct measure of overlap in product market offerings.

Following [Hsu et al. \(2022a\)](#) and [Hsu et al. \(2022b\)](#), I use the USPTO Trademark Case Files Dataset and distinguish product trademarks from marks associated primarily with marketing or branding activity. Product trademarks are classified according to the International (Nice) Classification, which contains 45 classes, including 34 classes of goods and 11 classes of services.

For each startup-year, I construct a vector over the 45 Nice Classes:

$$M_{it} = (M_{it1}, M_{it2}, \dots, M_{it45}),$$

where M_{itk} is the age-weighted number of active product trademarks held by startup i in class k in year t . Age weighting allows the measure to reflect the startup's contemporaneous product market position rather than its complete historical trademark portfolio.

Pairwise product market similarity is defined as the cosine similarity between the trademark

vectors of startups i and j :

$$PM_{ijt} = \frac{M'_{it}M_{jt}}{\sqrt{M'_{it}M_{it}}\sqrt{M'_{jt}M_{jt}}}.$$

Higher PM_{ijt} indicates greater overlap in the product categories in which the two startups commercialize their offerings. When two startups have active product trademarks but no overlap in Nice Classes, their similarity is set to zero. A startup-year with no active product trademark is represented by a zero vector and has zero product market similarity to other startups. The trademark data identify product market positions for 41.8% of startups in the sample.⁸

I aggregate pairwise product market similarity across other active startups to construct the focal startup's product market similarity:

$$\text{Product Market Similarity}_{it} = \sum_{j \neq i} PM_{ijt}.$$

Higher Product Market Similarity indicates greater overlap between the focal startup's commercialized offerings and those of other active startups.

For the heterogeneity analysis, both dimensions are evaluated at a predetermined baseline. *High Technological Similarity (High Tech)* equals one when predetermined Technological Similarity exceeds the median, and zero otherwise. *High Product Market Similarity (High PM)* is defined analogously using predetermined Product Market Similarity. For startups incorporated in states that never adopt a COW, market position is evaluated at the median adoption year of treated states.

4.4 Measuring Common VC Ownership

I define common VC ownership as the extent to which a startup is connected, through its VC investors, to rival startups. A simple count of shared-VC relationships would assign equal importance to connections with closely related rivals and with otherwise unrelated portfolio startups. I

⁸The Appendix examines trademark coverage directly. Appendix Table G1 compares startups with and without observable trademark activity. Table G2 and Figure G1 reconstruct the market-position analysis among startups with observable trademarks, and Table H1 restricts the sample to startups with both patent and trademark activity. The main patterns remain similar across these restricted samples.

therefore weight shared-VC connections by the degree of rivalry between the connected startups.

Similarity in business descriptions from PitchBook and VentureXpert provides a broad measure of pairwise rivalry. The descriptions summarize startups' products, services, and markets.⁹ I standardize the text by lower-casing, removing punctuation and stop words, and stemming the remaining terms, and then convert each description into a term frequency–inverse document frequency (TF-IDF) vector (Manning et al., 2008). Business-description similarity, BUS_{ij} , is the cosine similarity between the vectors of startups i and j , with higher values indicating greater overlap in their business activities.

Because business descriptions are observed as a single snapshot, I complement this static measure with patent-based technological similarity. The technological component captures variation in the relationship between startups as their technological activities evolve. To prevent technological convergence resulting from common VC ownership from mechanically entering the measure, technological similarity is fixed before the shared-VC relationship forms.

Let $SharedVC_{ijt}$ equal one if startups i and j share at least one VC investor in year t , and zero otherwise. Let τ_{ij} denote the last year in which startups i and j coexist without sharing a VC investor. Pre-connection technological similarity is defined as

$$TEC_{ij}^{pre} \equiv TEC_{ij\tau_{ij}}.$$

The two components of common VC ownership are

$$COBUS_{it} = \frac{\sum_{j \neq i} SharedVC_{ijt} \cdot BUS_{ij}}{\sum_{j \neq i} SharedVC_{ijt}} \times 100,$$

⁹I use business descriptions to identify economically relevant startups when constructing *CommonVC* because business descriptions are available for every startup in the sample and provide information about the businesses in which they operate. Using trademarks for this purpose would substantially reduce coverage, as many startups do not have products or register trademarks, even though they may still be economically relevant peers connected through common VC investors. Business descriptions are therefore better suited to defining the set of relevant startup relationships underlying *CommonVC*. However, because these descriptions are observed as a snapshot and do not vary over time, they are less suitable for measuring a startup's evolving product market position. Trademark records, by contrast, are dated and identify the specific goods and services through which startups commercialize their activities.

$$COTEC_{it} = \frac{\sum_{j \neq i} SharedVC_{ijt} \cdot TEC_{ij}^{pre}}{\sum_{j \neq i} SharedVC_{ijt}} \times 100.$$

$COBUS_{it}$ captures the focal startup's common-VC connections to startups operating in related businesses, while $COTEC_{it}$ captures connections to startups with related technological activities. Both components range from 0 to 100 and are set to zero when the startup has no shared-VC connection in a given year.

The main measure combines the two components:

$$CommonVC_{it} = \frac{COBUS_{it} + COTEC_{it}}{2}.$$

Higher CommonVC indicates that a startup is more strongly connected, through common VC investors, to economically relevant rival startups. Because the measure averages rather than sums across shared-VC connections, it captures the relevance of those connections rather than their raw number.¹⁰

4.5 Exposure to Knowledge Leakage

I measure the focal startup's exposure to knowledge leakage using geographic proximity and industry-level IP protection. Geographic proximity is captured by $VC-Peer\ Same\ CBSA_{it}$, which equals one when a common VC investor is located in the same Core-Based Statistical Area (CBSA) as at least one rival startup it also backs, and zero otherwise. Because CBSAs delineate economically integrated local areas based on commuting patterns, they provide a natural geographic unit for capturing settings in which local interaction and informal information exchange are more likely.

Industry-level IP protection is measured using the 2008 *Berkeley Patent Survey* (Graham et al., 2009). Biotechnology, medical devices, and information technology (IT) hardware are classified as strong-IP industries, while software and internet services are classified as weak-IP industries. $Strong\ IP\ Protection_{it}$ equals one for startups operating in strong-IP industries and zero for startups

¹⁰Appendix Tables D1–D4 replicate the main baseline and boundary condition results using $MALCommonVC$, which replaces Jaffe technological similarity with Mahalanobis technological similarity. The main results remain similar.

operating in weak-IP industries.

4.6 Director Interlocks

I construct *Director Interlock_{it}* to capture governance connections among startups linked through common VC investors. The measure equals one when a representative of a common VC investor sits on the boards of both the focal startup and another startup connected through the same investor, and zero otherwise.

4.7 Controls

The regressions include controls for both VC investor characteristics and the startup’s financing history. Investor characteristics are averaged across the startup’s prior VC investors and include VC age, number of funds, total prior startup investment rounds, and total IPOs among prior portfolio companies (Hsu, 2004; Guzman and Stern, 2020). These variables account for differences in investor experience, scale, and track record that may independently affect startup innovation.

Startup-level controls include the number of prior financing rounds and cumulative prior investment. These variables account for differences in financing history and resource availability that may also affect innovation.

4.8 Summary Statistics

Table 3 reports summary statistics at the startup-year level. CommonVC has a mean of 3.06 and exhibits substantial variation across startups. The average startup-year produces 0.67 patents and receives 4.64 five-year forward citations. Technological similarity also varies substantially across startups. The trademark-based Product Market Similarity has a mean of 0.06 and a median of zero. A startup-year with no active product mark holds no position in the trademark space and is recorded as zero; 32.5 percent of startup-years have a positive value. Cross-citations within the shared-VC portfolio are rare events. Only 2.3 percent of startup-years have a nonzero value of *Cross Cite*, so the variable’s percentiles are zero throughout. Finally, approximately half of startup-years have a common VC investor co-located with at least one rival startup.

Insert Table 3 about here.

5 Empirical Strategy

5.1 Identification and Estimation Framework

Estimating the causal effect of common VC ownership on startup innovation raises endogeneity concerns because common VC relationships are not formed randomly. More capable VC investors may systematically select more innovative startups, while more innovative startups may also be more likely to attract investment from overlapping sets of VC investors (Kortum and Lerner, 2001; Gompers et al., 2009; Sørensen, 2007). The observed relationship between common VC ownership and innovation may therefore reflect investor selection and startup characteristics rather than the effect of common VC ownership itself.

I address these concerns by exploiting the staggered adoption of Corporate Opportunity Waivers (COWs) as a legal shock to common VC ownership. As discussed in Section 3, COW adoption reduced the legal risk associated with VC investors holding stakes in competing startups, thereby making it easier for a VC investor to invest simultaneously in multiple competing startups. Because the legal change originates at the incorporation-state level rather than from the innovation performance of individual startups or the investment decisions of particular VC investors, it provides plausibly exogenous variation in common VC ownership.¹¹

The empirical design compares changes among startups incorporated in states that adopt COWs with contemporaneous changes among startups incorporated in states that have not yet adopted a COW. All specifications include startup fixed effects, headquarters-state-by-year fixed effects, and industry-by-year fixed effects, which absorb time-invariant startup characteristics and time-varying regional and industry shocks. Standard errors are clustered by state of incorporation.

I use COW adoption as an instrument for common VC ownership in a two-stage least squares

¹¹Appendix Figure J1 further examines whether COW adoption is associated with changes in VC investor characteristics or investment behavior. Using exposure to Delaware’s 2000 adoption based on investors’ pre-adoption portfolio composition, I find little evidence of changes in fundraising, investment scale, or broad investment style around adoption.

(2SLS) framework and, where applicable, report the corresponding reduced-form estimates alongside the instrumental variable (IV) results. The IV estimates capture the effect of COW-induced variation in common VC ownership, while the reduced-form estimates capture the effect of COW adoption directly on the corresponding outcome.

5.1.1 First Stage

The first stage estimates the effect of COW adoption on common VC ownership:

$$\text{CommonVC}_{it} = \alpha + \pi(\text{Treat}_{it} \times \text{Post}_{it}) + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \varepsilon_{it}, \quad (1)$$

where CommonVC_{it} measures the extent to which startup i is connected through common VC investors to economically relevant rival startups. $\text{Treat}_{it} \times \text{Post}_{it}$ equals one after the state in which startup i is incorporated adopts a COW. X_{it} contains the VC investor and startup financing controls described above, f_i denotes startup fixed effects, γ_{ht} denotes headquarters-state-by-year fixed effects, and ρ_{nt} denotes industry-by-year fixed effects. Because CommonVC is a continuous measure, π captures the change in the intensity of common VC ownership induced by COW adoption rather than a change in a binary common-ownership status.

 Insert Figure 1 about here.

Figure 1 reports the corresponding event-study estimates. The pre-adoption coefficients are statistically indistinguishable from zero, providing little evidence of differential trends in common VC ownership prior to COW adoption. Following adoption, common VC ownership increases and remains persistently higher in subsequent years.

5.1.2 Startup Innovation

The second stage estimates the effect of COW-induced variation in common VC ownership on startup innovation:

$$\text{Innovation}_{it} = \alpha + \beta \widehat{\text{CommonVC}}_{it} + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \varepsilon_{it}, \quad (2)$$

where $\widehat{\text{CommonVC}}_{it}$ is the fitted value from Equation (1). The dependent variables are Patent Count, 5-Year Citation, and 5-Year Citation (NoSelf). The coefficient β captures the effect of COW-induced variation in common VC ownership on startup innovation. The corresponding reduced-form specification replaces $\widehat{\text{CommonVC}}_{it}$ with $\text{Treat}_{it} \times \text{Post}_{it}$.

5.1.3 Technological Similarity and Patent Cross-Citations

To assess whether the innovation response is accompanied by patterns consistent with knowledge facilitation, I examine technological connections among startups. If common VC ownership facilitates access to relevant knowledge, COW-induced increases in common VC ownership should be associated with closer technological relationships.

I apply the same IV and reduced-form framework in Equations (1) and (2) using three measures of technological connection: *Technological Similarity*, *Technological Similarity (Mahalanobis)*, and *Cross Cite*. Higher values indicate closer technological relationships between the focal startup and other startups.

5.1.4 Director Interlocks as a Governance Channel

Director interlocks provide an observable governance channel through which common VC investors can remain directly involved across multiple portfolio startups. A director interlock arises when a representative of a common VC investor serves on the boards of both the focal startup and another startup connected through that investor.

I conduct two related tests. First, I examine whether common VC ownership is associated with stronger board-level connections by applying the same IV and reduced-form framework with Director Interlock as the dependent variable. This specification tests whether COW-induced increases in common VC ownership are accompanied by a greater likelihood of director interlocks among startups connected through common VC investors.

Second, I examine whether the innovation response to COW adoption varies with preexisting board-level connections. If direct investor involvement is an important channel through which common VC ownership affects innovation, the response should be stronger when such connections are already in place. I therefore return to the innovation outcomes and estimate the reduced-form

response separately by predetermined interlock status:

$$\begin{aligned} \text{Innovation}_{it} = & \alpha + \beta_1 [(\text{Treat}_{it} \times \text{Post}_{it}) \times \text{Interlock}_i] \\ & + \beta_2 [(\text{Treat}_{it} \times \text{Post}_{it}) \times (1 - \text{Interlock}_i)] \\ & + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \varepsilon_{it}, \end{aligned} \quad (3)$$

where Interlock_i equals one if startup i had a director interlock in the last observed panel year at or before its COW adoption cohort, and zero otherwise. For startups incorporated in states that never adopt a COW, interlock status is evaluated at the median adoption year of treated states. Measuring interlock status before the adoption of COWs avoids conditioning on board connections that may themselves respond to the legal shock. Because the two groups are mutually exclusive and exhaustive, the interaction terms already sum to $\text{Treat}_{it} \times \text{Post}_{it}$, and the specification omits the uninteracted $\text{Treat}_{it} \times \text{Post}_{it}$ term. β_1 and β_2 capture the reduced-form response to COW adoption for startups with and without a predetermined director interlock, respectively.

5.2 Boundary Conditions

The theory identifies two sets of boundary conditions that shape the balance between knowledge facilitation and knowledge dampening. The first concerns the focal startup's exposure to knowledge leakage, captured by geographic proximity and IP protection. The second concerns its market position, characterized by technological similarity and product market similarity.

5.2.1 Exposure to Knowledge Leakage

I first examine whether the effect of common VC ownership varies with exposure to knowledge leakage. As developed in Section 2.2.1, geographic proximity captures the likelihood that information disclosed to a common VC investor reaches a rival startup, whereas IP protection captures the expected consequences if such leakage occurs. I measure these dimensions using VC–Peer Same CBSA and Strong IP Protection, respectively.

For each moderator M_{it} , both CommonVC_{it} and its interaction with M_{it} are treated as endogenous. They are instrumented with COW adoption and the interaction between COW adoption and

the moderator. The two first-stage equations are

$$\begin{aligned} \text{CommonVC}_{it} = & \alpha + \pi_1(\text{Treat}_{it} \times \text{Post}_{it}) + \pi_2(\text{Treat}_{it} \times \text{Post}_{it} \times M_{it}) \\ & + \delta M_{it} + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \varepsilon_{it}, \end{aligned} \quad (4)$$

$$\begin{aligned} \text{CommonVC}_{it} \times M_{it} = & \alpha + \lambda_1(\text{Treat}_{it} \times \text{Post}_{it}) + \lambda_2(\text{Treat}_{it} \times \text{Post}_{it} \times M_{it}) \\ & + \eta M_{it} + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \nu_{it}. \end{aligned} \quad (5)$$

The corresponding second stage is

$$\begin{aligned} \text{Innovation}_{it} = & \alpha + \beta_1 \widehat{\text{CommonVC}}_{it} + \beta_2 (\widehat{\text{CommonVC}} \times M)_{it} + \delta M_{it} \\ & + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \varepsilon_{it}. \end{aligned} \quad (6)$$

I estimate this specification separately for VC–Peer Same CBSA and Strong IP Protection. The coefficient β_2 captures how that effect varies with the corresponding dimension of exposure to knowledge leakage.

5.2.2 Market Position

The second set of boundary conditions concerns the focal startup’s market position. Section 2.2.2 predicts that the innovation consequences of common VC ownership depend jointly on technological similarity and product market similarity. As described in Section 4.3, I measure both dimensions using predetermined values and classify startups as High Tech or Low Tech and High PM or Low PM according to whether their technological similarity and product market similarity lie above or below the corresponding sample medians. Combining these two dimensions yields four mutually exclusive market positions: High Tech/Low PM, High Tech/High PM, Low Tech/Low PM, and Low Tech/High PM.¹²

For this analysis, I estimate the reduced-form response to COW adoption across the four market positions:

¹²Appendix Table F1 reports the composition and treatment support of the four market-position cells, and Appendix Figure F1 shows their distribution in the sample.

$$\begin{aligned} \text{Innovation}_{it} = & \alpha + \sum_{k=1}^4 \beta_k \left[(\text{Treat}_{it} \times \text{Post}_{it}) \times G_i^k \right] \\ & + X_{it} + f_i + \gamma_{ht} + \rho_{nt} + \varepsilon_{it}, \end{aligned} \quad (7)$$

where G_i^k denotes the four predetermined market-position groups. The coefficients β_k capture the innovation response to COW adoption within each market position. Because market position is measured before COW adoption, the specification does not condition on technological or product market similarity that may itself respond to the legal shock. Equation (7) includes all four group interactions and drops the uninteracted $\text{Treat}_{it} \times \text{Post}_{it}$ term. Because the four groups are mutually exclusive and exhaustive within the estimation sample, the interactions already sum to that term, and retaining it would introduce perfect collinearity. In this specification, no market position serves as an omitted comparison category. It reports each β_k as the group-specific reduced-form response rather than as a difference from an excluded group. Comparing the coefficients across the four groups therefore provides evidence on how the innovation response varies with the market positions developed in the theory.

6 Results

6.1 Baseline Effect of Common VC Ownership

Table 4 reports the baseline estimates. Column (1) of Panel A shows that COW adoption increases CommonVC by 0.665 in the first stage, with a first-stage F -statistic of 49.09 that alleviates concerns about a weak IV. Relative to the sample mean of 3.06, this estimate corresponds to an increase of approximately 22% in common VC ownership.

Columns (2)–(4) report the second-stage estimates. The coefficient on CommonVC is 0.104 for Patent Count, 0.195 for 5-Year Citation, and 0.185 for 5-Year Citation (NoSelf). Because the innovation outcomes are measured as $\ln(1 + y)$, these estimates correspond to increases of approximately 11%, 22%, and 20%, respectively. Panel B reports positive and statistically significant reduced-form estimates of 0.069, 0.130, and 0.123 across the three outcomes. Although smaller in

magnitude because they capture the effect of COW adoption without scaling by the first stage, the reduced-form estimates exhibit a similar pattern to the IV estimates.

The positive effects on both patent counts and subsequent citations are informative about the balance between knowledge facilitation and knowledge dampening. Knowledge dampening may induce startups to accumulate patents defensively to protect against appropriation by rival portfolio startups (Ziedonis, 2004). Such behavior could increase patent counts without a corresponding increase in the subsequent technological impact of those patents. The increase in both 5-Year Citation and 5-Year Citation (NoSelf) therefore weighs against an interpretation based primarily on defensive patent accumulation. Overall, the baseline results indicate that the innovation benefits associated with knowledge facilitation outweigh knowledge dampening on average.

Insert Table 4 about here.

The baseline estimates are robust across a range of alternative samples, measures, and placebo tests. Because Delaware-incorporated startups account for a large share of the sample, I exclude them to assess whether the results are disproportionately driven by Delaware. I also restrict the sample to states with substantial entrepreneurial activity to ensure that the findings are not driven by relatively thin startup markets and, separately, exclude Silicon Valley startups to assess whether they depend on the distinctive concentration of venture activity in Silicon Valley. Across these alternative samples, the estimated effects remain positive and statistically significant. The results are also robust to replacing the Jaffe-based technological component of CommonVC with its Mahalanobis-based alternative. Finally, in a placebo test, COW adoption has no significant effect on either CommonVC or innovation among startups unlikely to face a plausible rival, while both effects are present among startups with a plausible rival.¹³

¹³Appendix Table A2 excludes Delaware-incorporated startups, Table B1 restricts the sample to states with substantial entrepreneurial activity, Tables C1 and C2 examine samples outside and within Silicon Valley, respectively, Tables D1–D4 report results using the Mahalanobis-based common VC measure, and Table E1 reports the no-plausible-rival placebo test.

6.2 Evidence on Knowledge Facilitation

6.2.1 Technological Similarity and Patent Cross-Citations

Table 5 provides additional evidence on knowledge facilitation. Columns (2) and (3) report the effects on Technological Similarity using the Jaffe and Mahalanobis measures, respectively. The second-stage coefficients are 0.263 and 0.253, both statistically significant at the 1% level. The corresponding reduced-form estimates in Panel B are 0.175 and 0.168, respectively, and are also statistically significant at the 1% level. Thus, COW-induced increases in common VC ownership are associated with greater technological similarity.

Column (4) examines patent cross-citations among startups connected through common VC investors. The second-stage coefficient on *CommonVC* is 0.041, while the corresponding reduced-form estimate is 0.027; both are statistically significant at the 1% level. COW-induced increases in common VC ownership are therefore accompanied by greater technological referencing within the shared-VC network.

Taken together, the increases in technological similarity and patent cross-citations provide complementary evidence consistent with knowledge facilitation. These outcomes do not directly measure the underlying knowledge flows, but they show that greater common VC ownership is accompanied by closer technological linkages among startups.

Insert Table 5 about here.

6.2.2 Director Interlocks as a Governance Channel

If direct investor involvement is an important channel through which common VC ownership affects innovation, the effect should be stronger when the common VC investor is more directly involved in its portfolio startups. Director interlocks provide an observable governance channel through which such involvement can occur. Table 6 examines this channel in two ways. Panel A tests whether COW-induced increases in common VC ownership are accompanied by greater director interlocking among startups connected through common VC investors. Panel B examines

whether the innovation response to COW adoption is stronger when such a board-level connection was already in place.

Panel A shows that COW-induced increases in common VC ownership are accompanied by greater director interlocking. Column (1) reports the reduced-form estimate. COW adoption increases the probability of a director interlock by 0.87 percentage points, relative to a sample mean of 6.47%. Columns (2) and (3) report the corresponding first- and second-stage estimates. The second-stage coefficient on *CommonVC* is 0.0131 and statistically significant at the 1% level.

Panel B examines whether the innovation response to COW adoption varies with interlock status fixed before adoption. Across all three innovation outcomes, the response is substantially larger among startups with an interlock already in place. For patent counts, the coefficient is 0.231 among startups with an interlock and 0.068 among those without one. The corresponding estimates are 0.355 and 0.128 for five-year citations and 0.426 and 0.121 for five-year non-self citations. The innovation response is therefore roughly three times as large among startups with a predetermined interlock.

Taken together, the two panels provide complementary evidence that director interlocks are an observable governance channel associated with common VC ownership. COW-induced increases in common VC ownership strengthen board-level connections across portfolio startups, and the innovation response to COW adoption is substantially larger when such connections were already in place. This pattern is consistent with board representation allowing common VC investors to remain engaged across multiple portfolio startups.

Insert Table 6 about here.

6.3 Exposure to Knowledge Leakage

I examine whether the innovation effects of common VC ownership vary with the focal startup's exposure to knowledge leakage. Following Section 2.2.1, I consider two dimensions of such exposure: geographic proximity and IP protection.

6.3.1 Geographic Proximity

Table 7 examines whether the focal startup realizes smaller innovation gains from common VC ownership when its common VC investor is geographically close to a rival startup in the same portfolio. The interaction between CommonVC and VC–Peer Same CBSA is negative across all three innovation outcomes, with coefficients of -0.175 for Patent Count, -0.318 for 5-Year Citation, and -0.320 for 5-Year Citation (NoSelf). All three estimates are statistically significant at the 1% level.

The magnitudes indicate substantial attenuation. When VC–Peer Same CBSA=0, the coefficients on CommonVC are 0.271, 0.495, and 0.483 across the three outcomes. When the common VC investor is co-located with a rival of the focal startup, the corresponding implied coefficients fall to 0.096, 0.177, and 0.163. Thus, the focal startup realizes substantially smaller innovation gains from common VC ownership when its common VC investor is geographically close to a rival it also backs. This pattern is consistent with the theory that greater exposure to potential knowledge leakage shifts the balance away from knowledge facilitation and toward knowledge dampening, as the focal startup becomes more cautious about sharing information with the common VC investor and devotes greater attention to protecting its proprietary knowledge.

Insert Table 7 about here.

6.3.2 IP Protection

Table 8 examines whether stronger IP protection amplifies the innovation benefits of common VC ownership. The interaction between CommonVC and Strong IP Protection is positive and statistically significant across all three innovation outcomes, with coefficients of 0.099 for Patent Count, 0.135 for 5-Year Citation, and 0.126 for 5-Year Citation (NoSelf).

The magnitudes indicate substantial amplification. In weak-IP industries, the coefficients on CommonVC are 0.049, 0.131, and 0.126 across the three outcomes. In strong-IP industries, the corresponding implied coefficients increase to 0.148, 0.266, and 0.252. The innovation gains from common VC ownership are therefore two to three times larger in industries with stronger IP pro-

tection than in those with weaker protection. This pattern is consistent with the theory that stronger appropriability reduces the costs associated with potential knowledge leakage, shifting the balance toward knowledge facilitation and away from knowledge dampening.

 Insert Table 8 about here.

Taken together, the geographic-proximity and IP-protection results show that the innovation benefits of common VC ownership vary systematically with exposure to knowledge leakage. The gains are smaller when the common VC investor is geographically close to a rival startup and larger when proprietary knowledge is more strongly protected, consistent with the predicted balance between knowledge facilitation and knowledge dampening.

6.4 Market Position

Finally, I examine heterogeneity across the four predetermined market positions developed in Section 2.2.2 and summarized in Table 1. Table 9 reports the corresponding reduced-form responses to COW adoption.

For High Tech/Low PM startups, in the *knowledge corridor* quadrant, the theory predicts the strongest positive effect because knowledge facilitation is strong while knowledge dampening remains relatively limited. Consistent with this prediction, COW adoption increases Patent Count by 0.178, 5-Year Citation by 0.338, and 5-Year Citation (NoSelf) by 0.317. These are the largest point estimates across the four market positions for all three innovation outcomes, with the patent effect about 10 percent higher and the citation effects about 50 percent higher than in the *coordinated rivals* quadrant.

Among High Tech/High PM startups, in the *coordinated rivals* quadrant, both knowledge facilitation and knowledge dampening are strong, making the net effect theoretically ambiguous. The estimated responses are nevertheless positive, at 0.162 for Patent Count, 0.225 for 5-Year Citation, and 0.213 for 5-Year Citation (NoSelf). Although positive, these point estimates are smaller than those for the High Tech/Low PM group. The empirical pattern therefore suggests that, on average,

the benefits of knowledge facilitation outweigh the costs of knowledge dampening in this market position, although by a smaller margin than in the *knowledge corridor*.

For Low Tech/Low PM startups, in the *faint tailwind* quadrant, both knowledge facilitation and knowledge dampening are relatively weak, so the theory predicts only a small net effect. Consistent with this prediction, the estimated responses are 0.025 for Patent Count, 0.096 for 5-Year Citation, and 0.085 for 5-Year Citation (NoSelf). Although significantly positive, these estimates are roughly one-seventh of the patent effect and less than one-third of the citation effects in the *knowledge corridor* quadrant.

For Low Tech/High PM startups, in the *contested market* quadrant, opportunities for knowledge facilitation are limited while incentives for knowledge dampening are strong, leading the theory to predict a negative net effect. The estimated responses are negative across all three outcomes: -0.022 for Patent Count, -0.052 for 5-Year Citation, and -0.046 for 5-Year Citation (NoSelf). The two citation estimates are statistically significant at the 5% level, while the patent-count estimate is not statistically distinguishable from zero.

Insert Table 9 about here.

7 Conclusion

This paper studies common VC ownership, in which a startup shares a VC investor with rival startups, and examines how such relationships shape startup innovation. I exploit the staggered adoption of COWs, which reduced legal barriers to VC investors holding stakes in competing startups, to isolate COW-induced variation in common VC ownership. The evidence shows that common VC ownership increases innovation on average. It raises both patenting and subsequent patent citations, including non-self citations, suggesting that the increase in patenting is accompanied by greater subsequent technological impact. Common VC ownership also increases technological similarity and, among connected startups, patent cross-citations, providing complementary evidence consistent with knowledge facilitation. Moreover, COW adoption increases director interlocks

among connected startups, providing an observable governance channel through which common VC investors can remain directly involved across multiple portfolio firms.

These innovation benefits, however, are not uniform. Their magnitude depends on the balance between opportunities for knowledge facilitation and exposure to potential knowledge leakage. The gains become substantially smaller when the common VC investor is geographically close to a rival startup it also backs, but become larger in industries with stronger IP protection. These patterns are consistent with greater exposure to potential knowledge leakage shifting the balance away from knowledge facilitation and toward knowledge dampening, while stronger protection of proprietary knowledge shifts the balance in the opposite direction.

The balance between knowledge facilitation and knowledge dampening also varies with the startup's market position. The largest positive innovation response occurs when technological similarity is high and product market similarity is low, a setting in which technological overlap creates substantial opportunities for knowledge facilitation while direct product market rivalry remains limited. When both technological similarity and product market similarity are high, the effect remains positive but is smaller, suggesting that knowledge facilitation still outweighs knowledge dampening on average even when both forces are strong. When both dimensions are low, the positive innovation response becomes substantially smaller. Finally, when technological similarity is low and product market similarity is high, the effect turns negative for citation-based measures of innovation, consistent with knowledge dampening becoming more important when opportunities for facilitation are limited and product market rivalry is strong. Together, these results show that the direction and magnitude of the innovation consequences of common VC ownership depend importantly on the technological and competitive relationships among portfolio startups.

The analysis also leaves several questions for future research. First, the innovation measures in this paper rely primarily on patents and forward citations. These outcomes capture an important dimension of technological innovation but do not observe innovations protected through secrecy, copyrights, or other mechanisms, nor do they fully capture product development that never results in patent applications. Product market similarity is measured using trademark data, which provide

an observable measure of firms' commercial positioning but may not capture the full range of products, services, or competitive overlap among startups. Future research could extend the analysis to nonpatented innovation, product introductions, and other measures of product market competition.

Second, the empirical analysis does not fully explore heterogeneity across common VC investors. Common VC investors may differ considerably in the expertise they possess and in how they engage with their portfolio firms, and these differences may shape the effect of common VC ownership on startup innovation. Examining investor domain expertise, partner backgrounds, technical personnel, prior portfolio experience, or other sources of investor knowledge could help identify more directly which types of common VC investors are more likely to generate stronger innovation effects across their portfolio firms.

Third, the analysis treats technological similarity and product market similarity primarily as predetermined dimensions of market position. Yet common VC investors may also shape these positions over time through investment selection, strategic advice, technological guidance, or product market decisions. Future research could therefore examine whether common VC ownership changes the subsequent technological and commercial positioning of portfolio startups and how such repositioning affects competition and innovation.

The findings offer distinct implications for VC investors and entrepreneurs. For VC investors, the results suggest that investing across competing startups can create innovation benefits when the investor possesses expertise that is relevant across firms, but those benefits may erode as product market rivalry and exposure to knowledge leakage increase. Portfolio construction should therefore account not only for technological relatedness across startups, but also for the competitive risks created by backing close rivals. For entrepreneurs, the findings suggest that sharing a VC investor with rival startups can be valuable when the investor's knowledge is highly relevant and direct product market competition is limited, but it can also create concerns about the protection of proprietary information. Entrepreneurs should therefore evaluate a prospective investor not only in terms of the resources and expertise it can provide, but also in terms of its relationships with competing portfolio firms and the safeguards surrounding sensitive knowledge.

More broadly, the findings may inform the ongoing debate over common ownership, which has focused largely on public firms and on concerns that common ownership may weaken competition and reduce incentives to innovate. The VC-backed startup setting differs in several important respects. VC investors often hold concentrated stakes, remain actively involved in portfolio-company governance, and pursue investment objectives centered on the value and eventual exit of a relatively small set of portfolio firms rather than on returns across a broadly diversified public-equity portfolio. In this setting, I find that common VC ownership increases innovation on average, but its effects depend importantly on the technological and competitive relationships among the firms connected through the common investor. For policymakers considering the antitrust implications of common ownership, this evidence may therefore provide a useful perspective on the conditions under which common ownership is associated with innovation gains and those under which competitive concerns become more salient.

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Table 1: Common VC Ownership and Startup Innovation, by Startup Market Position

Technological similarity	Product market similarity	
	Low	High
High	<i>Knowledge corridor</i> Facilitation: strong Dampening: weak Common VC: strong incentive to facilitate and mitigate concerns Expected effect: positive <i>Illustrative pair: CFS–SpaceX</i>	<i>Coordinated rivals</i> Facilitation: strong Dampening: strong Common VC: strong incentive to facilitate, but weak incentive to mitigate concerns Expected effect: ambiguous <i>Illustrative pair: CFS–Proxima Fusion</i>
Low	<i>Faint tailwind</i> Facilitation: weak Dampening: weak Common VC: weak incentive to facilitate or mitigate concerns Expected effect: small or null <i>Illustrative pair: CFS–Zipline</i>	<i>Contested market</i> Facilitation: weak Dampening: strong Common VC: weak incentive to facilitate or mitigate concerns Expected effect: negative <i>Illustrative pair: CFS–General Fusion</i>

Note: The table summarizes how startup market position shapes the balance between knowledge facilitation and knowledge dampening under common VC ownership. All illustrative pairs use Commonwealth Fusion Systems (CFS) as the focal startup. CFS and SpaceX share Brightwing Capital; CFS and Proxima Fusion share Brevan Howard Macro Venture Fund; and CFS shares Evolution VC Partners with Zipline and General Fusion.

Table 2: Summary of Corporate Opportunity Waiver Timeline

State incorp	Effective Date	By Charter	By Bylaws	By Board Action	Covers Directors	Covers Shareholders
DE	7/1/2000	Yes	Possible	Yes	Yes	Yes
OK	11/1/2001	Yes	Possible	Yes	Yes	Yes
MO	10/1/2003	Yes	Possible	Yes	Yes	Yes
KS	1/1/2005	Yes	Possible	Yes	Yes	Yes
TX	1/1/2006	Yes	Possible	Yes	Yes	Yes
NV	10/1/2007	Yes	Possible	Yes	Yes	Yes
NJ	3/11/2011	Yes	Possible	Yes	Yes	Yes
MD	10/1/2014	Yes	Possible	Yes	Yes	No
WA	1/1/2016	Yes	No	No	Yes	Yes

Notes: This table provides summary information on the states that amended their corporate laws explicitly to allow COWs (Source: [Rauterberg and Talley \(2017\)](#)).

Table 3: Summary Statistics

Variable	Obs.	Mean	Std. Dev.	p10	p50	p90
CommonVC	170,809	3.06	5.09	0.00	0.51	9.13
Patent Count	170,809	0.67	1.87	0.00	0.00	2.00
5-Yr Cit.	170,809	4.64	17.65	0.00	0.00	9.00
5-Yr Cit. (NS)	170,809	3.67	13.90	0.00	0.00	7.00
Tech. Sim.	170,809	3.14	2.46	0.00	3.50	6.33
Tech. Sim. (Mal.)	170,809	3.41	2.54	0.00	4.01	6.50
Product Mkt. Sim.	170,809	0.06	0.11	0.00	0.00	0.28
Cross Cite	170,809	0.05	0.34	0.00	0.00	0.00
VC–Peer Same CBSA	170,809	0.49	0.50	0.00	0.00	1.00
Strong IP	118,117	0.67	0.47	0.00	1.00	1.00
VC Age	170,809	6.51	9.51	0.00	0.00	21.00
# VC Funds	170,809	2.00	3.51	0.00	0.00	6.71
VC Total Investment Rounds	170,809	60.81	123.29	0.00	0.00	210.00
VC Total IPOs	170,809	12.08	26.14	0.00	0.00	43.17
# VC Deals	170,809	1.59	2.72	0.00	0.00	5.00
Deal Value	170,809	4.27	18.15	0.00	0.00	8.19

Notes: This table reports summary statistics at the startup-year level. “5-Yr Cit. (NS)” denotes five-year forward citations excluding self-citations; “Tech. Sim.” denotes Technological Similarity; “Mal.” denotes the Mahalanobis-based measure; and “Strong IP” denotes Strong IP Protection. “Product Mkt. Sim.” is the trademark-based Product Market Similarity: the cosine similarity between the startup’s age-weighted Nice-class trademark vector and those of its peers. A startup-year with no active product mark has no position in the trademark space and is recorded as zero; 32.5 percent of startup-years have a positive value. “Cross Cite” counts patent citations exchanged with shared-VC peers; about 2.3 percent of startup-years have a nonzero within-VC-portfolio cross-citation, which is why its percentiles are zero throughout.

Table 4: Common VC Ownership and Startup Innovation

	First Stage	Second Stage		
	(1)	(2)	(3)	(4)
	CommonVC	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	0.665*** (0.095)			
$\widehat{\text{CommonVC}}$		0.104*** (0.019)	0.195*** (0.045)	0.185*** (0.039)
Observations	168,702	168,702	168,702	168,702
KP Wald F	49.09			
Adjusted R^2	0.660			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.069*** (0.011)	0.130*** (0.024)	0.123*** (0.021)
Observations		168,702	168,702	168,702
Adjusted R^2		0.530	0.470	0.460
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: Panel A reports IV estimates. Column (1) reports the first stage, and Columns (2)–(4) report the second-stage estimates. Panel B reports the corresponding reduced-form estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Common VC Ownership, Technological Similarity, and Patent Cross-Citations

	First Stage	Second Stage		
	(1)	(2)	(3)	(4)
	CommonVC	Tech. Sim.	Tech. Sim. (Mal.)	Cross Cite
Panel A: IV Estimates				
Treat $\times$ Post	0.665*** (0.095)			
$\widehat{\text{CommonVC}}$		0.263*** (0.055)	0.253*** (0.059)	0.041*** (0.009)
Observations	168,702	168,702	168,702	168,702
KP Wald F	49.09			
Adjusted R^2	0.660			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.175*** (0.045)	0.168*** (0.048)	0.027*** (0.004)
Observations		168,702	168,702	168,702
Adjusted R^2		0.674	0.662	0.361
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: “Tech. Sim.” denotes *Technological Similarity*, and “Mal.” denotes the Mahalanobis-based alternative. *Cross Cite* measures patent cross-citations between the focal startup and startups connected through common VC investors. Panel A reports IV estimates, and Panel B reports the corresponding reduced-form estimates. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Common VC Ownership and Director Interlocks

Panel A: Does COW Adoption Raise Director Interlocks?			
	Reduced Form	IV	
	(1)	(2)	(3)
	Director Interlock	First Stage CommonVC	Second Stage Director Interlock
Treat $\times$ Post	0.0087*** (0.0025)	0.665*** (0.095)	
$\widehat{\text{CommonVC}}$			0.0131*** (0.0039)
Mean Dep. Var.	0.0647		0.0647
Observations	168,702	168,702	168,702
KP Wald F		49.09	
Adjusted R^2	0.401	0.660	
Panel B: Innovation Response by Predetermined Interlock Status			
	(1)	(2)	(3)
	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Treat $\times$ Post $\times$ Interlock	0.231*** (0.073)	0.355*** (0.107)	0.426*** (0.091)
Treat $\times$ Post $\times$ No Interlock	0.068*** (0.011)	0.128*** (0.024)	0.121*** (0.022)
Observations	168,702	168,702	168,702
Adjusted R^2	0.453	0.392	0.376
Other Controls	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes

Notes: A director interlock equals one when a representative of a common VC investor sits on the boards of both the focal startup and another startup connected through the same investor. In Panel A, Column (1) reports the reduced-form estimate. Columns (2) and (3) report the first and second stages of the IV specification. In Panel B, interlock status is predetermined using the last observed panel year at or before the relevant COW adoption cohort. The specification includes mutually exclusive interactions of Treat $\times$ Post with indicators for preexisting interlock and no preexisting interlock. The reported coefficients therefore represent the reduced-form response to COW adoption within each group. The dependent variables are the natural logarithm of one plus patent counts, five-year forward citations, and five-year non-self citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Common VC Ownership, Geographic Co-Location, and Startup Innovation

	First Stage		Second Stage		
	(1) CommonVC	(2) CommonVC $\times$ Same CBSA	(3) Patent Count	(4) 5-Yr Cit.	(5) 5-Yr Cit. (NS)
Treat $\times$ Post	−0.043 (0.087)	−0.316*** (0.077)			
Treat $\times$ Post $\times$ Same CBSA	1.077*** (0.269)	1.366*** (0.247)			
Same CBSA	3.430*** (0.320)	4.299*** (0.308)	−0.174 (0.145)	−0.286 (0.264)	−0.238 (0.244)
$\widehat{\text{CommonVC}}$			0.271*** (0.066)	0.495*** (0.136)	0.483*** (0.128)
$\widehat{\text{CommonVC}} \times \text{Same CBSA}$			−0.175*** (0.063)	−0.318*** (0.114)	−0.320*** (0.110)
Observations	168,702	168,702	168,702	168,702	168,702
KP Wald F	13.21				
Other Controls	Yes	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Notes: “Same CBSA” denotes *VC–Peer Same CBSA*. Columns (1) and (2) report the first stages for *CommonVC* and its interaction with co-location. Columns (3)–(5) report the corresponding second-stage estimates. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: Common VC Ownership, IP Protection, and Startup Innovation

	First Stage		Second Stage		
	(1) CommonVC	(2) CommonVC $\times$ Strong IP	(3) Patent Count	(4) 5-Yr Cit.	(5) 5-Yr Cit. (NS)
Treat $\times$ Post	1.736*** (0.296)	-0.259* (0.138)			
Treat $\times$ Post $\times$ Strong IP	-1.223*** (0.301)	1.210*** (0.241)			
$\widehat{\text{CommonVC}}$			0.049*** (0.017)	0.131** (0.051)	0.126** (0.049)
$\widehat{\text{CommonVC}} \times \text{Strong IP}$			0.099*** (0.023)	0.135*** (0.050)	0.126** (0.052)
Observations	110,027	110,027	110,027	110,027	110,027
KP Wald F	19.59				
Other Controls	Yes	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

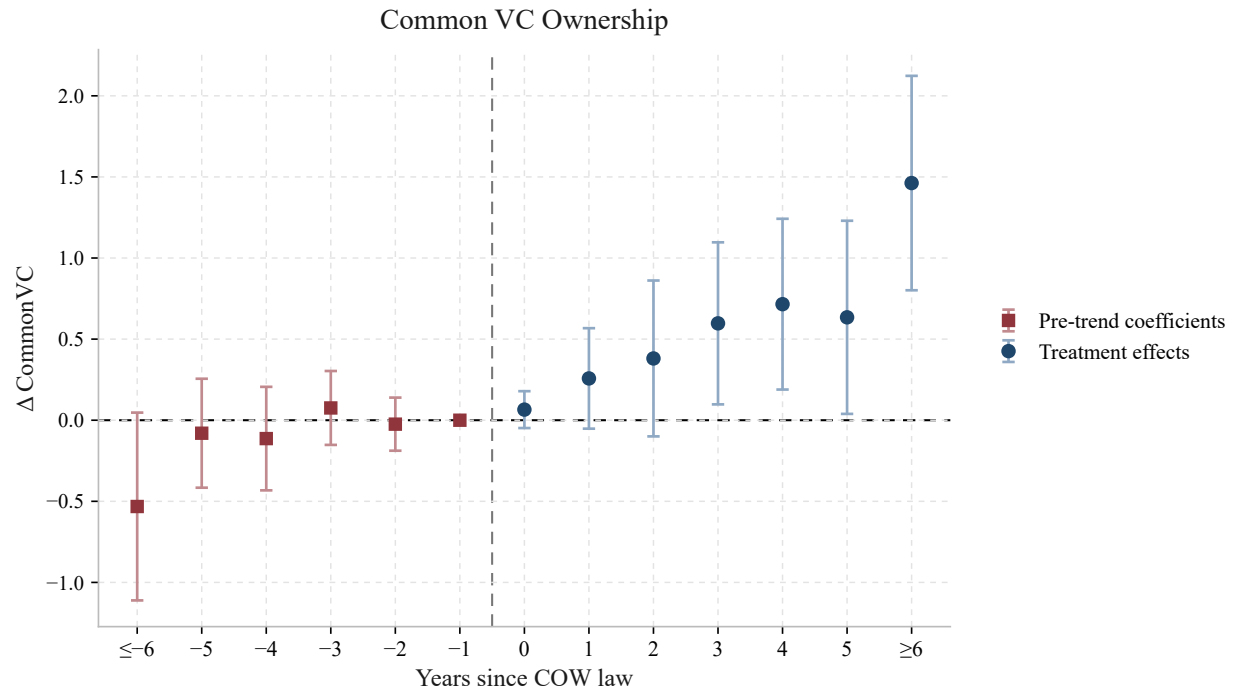
Notes: “Strong IP” denotes *Strong IP Protection*. Biotechnology, medical devices, and IT hardware are classified as strong-IP industries, while software and internet services are classified as weak-IP industries (Graham et al., 2009). Columns (1) and (2) report the first stages for *CommonVC* and its interaction with Strong IP Protection. Columns (3)–(5) report the corresponding second-stage estimates. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: COW Adoption and Innovation by Predetermined Market Position

	(1) Patent Count	(2) 5-Yr Cit.	(3) 5-Yr Cit. (NS)
Treat $\times$ Post $\times$ High Tech / Low PM	0.178*** (0.009)	0.338*** (0.022)	0.317*** (0.019)
Treat $\times$ Post $\times$ High Tech / High PM	0.162*** (0.021)	0.225*** (0.025)	0.213*** (0.022)
Treat $\times$ Post $\times$ Low Tech / Low PM	0.025** (0.010)	0.096*** (0.023)	0.085*** (0.020)
Treat $\times$ Post $\times$ Low Tech / High PM	−0.022 (0.014)	−0.052** (0.023)	−0.046** (0.022)
Mean Dep. Var.	0.257	0.487	0.452
Observations	86,824	86,824	86,824
Adjusted R^2	0.485	0.422	0.409
Other Controls	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes

Notes: This table reports reduced-form estimates of the effect of COW adoption on innovation across four predetermined market positions. Each reported coefficient corresponds to the interaction between *Treat* $\times$ *Post* and the indicated market-position group. “Tech” refers to predetermined *Technological Similarity*, “PM” refers to predetermined *Product Market Similarity*, and “NS” denotes citations excluding self-citations. The four market-position groups are mutually exclusive and exhaustive. Innovation outcomes are measured as $\ln(1 + y)$. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 1: The Impact of COWs on Common VC Ownership



Notes: This figure reports event-study estimates of the effect of COW adoption on common VC ownership. The omitted period is the year before adoption.

Appendix

A Excluding Delaware-Incorporated Startups

Delaware-incorporated startups account for approximately 66% of the sample (Table A1), raising the possibility that the baseline estimates are disproportionately influenced by startups incorporated in Delaware. I examine this concern by re-estimating the baseline specifications after excluding all Delaware-incorporated startups.

Table A1: Startup Incorporation State

State of Incorporation	Count	Share
DE	11,407	0.66
CA	1,224	0.07
NY	346	0.02
TX	325	0.02
PA	286	0.02
OH	275	0.02
MA	260	0.02
IL	244	0.01
MI	223	0.01
FL	213	0.01
MN	205	0.01

Notes: This table reports the number and share of startups by state of incorporation.

Table A2 shows that the baseline findings are not driven solely by Delaware-incorporated startups. COW adoption continues to increase *CommonVC*, with a Kleibergen–Paap Wald F -statistic of 12.03. The second-stage coefficients are positive for all three innovation outcomes: 0.082 for *Patent Count*, 0.114 for *5-Year Citation*, and 0.106 for *5-Year Citation (NoSelf)*. Panel B reports positive reduced-form estimates as well. Although the second-stage estimates are somewhat smaller than those in the full sample, their signs and overall pattern are consistent with the baseline results.

Table A2: Common VC Ownership and Startup Innovation: Excluding Delaware-Incorporated Startups

	First Stage	Second Stage		
	(1)	(2)	(3)	(4)
	CommonVC	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	2.107*** (0.608)			
$\widehat{\text{CommonVC}}$		0.082** (0.031)	0.114* (0.061)	0.106* (0.055)
Observations	57,357	57,357	57,357	57,357
KP Wald F	12.03			
Adjusted R^2	0.64			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.173*** (0.046)	0.240** (0.101)	0.224** (0.093)
Observations		57,357	57,357	57,357
Adjusted R^2		0.57	0.52	0.51
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: The sample excludes startups incorporated in Delaware. Panel A reports IV estimates. Column (1) reports the first stage, and Columns (2)–(4) report the second-stage estimates. Panel B reports the corresponding reduced-form estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Startups Incorporated in States with Substantial Entrepreneurial Activity

I restrict the sample to startups incorporated in Delaware, Massachusetts, New York, and California. This exercise examines whether the baseline findings continue to hold within states that account for a substantial share of U.S. entrepreneurial activity.

Table B1: Common VC Ownership and Startup Innovation: DE, MA, NY, and CA

	First Stage	Second Stage		
	(1) CommonVC	(2) Patent Count	(3) 5-Yr Cit.	(4) 5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	0.681*** (0.140)			
$\widehat{\text{CommonVC}}$		0.133*** (0.019)	0.277*** (0.035)	0.261*** (0.030)
Observations	130,523	130,523	130,523	130,523
KP Wald F	23.74			
Adjusted R^2	0.66			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.090** (0.022)	0.189*** (0.026)	0.178*** (0.028)
Observations		130,523	130,523	130,523
Adjusted R^2		0.53	0.47	0.46
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: The sample is restricted to startups incorporated in Delaware, Massachusetts, New York, and California. Panel A reports IV estimates and Panel B reports reduced-form estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B1 shows that the baseline pattern remains in this restricted sample. COW adoption increases *CommonVC*, and the second-stage estimates are positive and statistically significant for patent counts and both citation outcomes. The reduced-form estimates in Panel B show the same

pattern. The baseline findings therefore do not depend on including startups incorporated in states with relatively limited entrepreneurial activity.

C Startups Outside Silicon Valley

Because Silicon Valley accounts for a substantial share of U.S. venture activity, the baseline estimates could disproportionately reflect startups located in this region. I examine this possibility by estimating the baseline specifications separately outside and within Silicon Valley.

Table C1: Common VC Ownership and Startup Innovation: Excluding Silicon Valley

	First Stage	Second Stage		
	(1)	(2)	(3)	(4)
	CommonVC	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	0.723*** (0.091)			
$\widehat{\text{CommonVC}}$		0.086*** (0.014)	0.159*** (0.028)	0.151*** (0.025)
Observations	150,135	150,135	150,135	150,135
KP Wald F	63.89			
Adjusted R^2	0.65			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.062*** (0.012)	0.115*** (0.022)	0.109*** (0.021)
Observations		150,135	150,135	150,135
Adjusted R^2		0.52	0.47	0.46
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: The sample excludes startups located in the Silicon Valley area. Panel A reports IV estimates and Panel B reports reduced-form estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Tables C1 and C2 show that the positive relationship identified in the baseline analysis is present both outside and within Silicon Valley. Excluding Silicon Valley, COW adoption continues to increase *CommonVC*, and the second-stage estimates remain positive and statistically significant

Table C2: Common VC Ownership and Startup Innovation: Silicon Valley Only

	First Stage	Second Stage		
	(1)	(2)	(3)	(4)
	CommonVC	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	0.833*** (0.143)			
$\widehat{\text{CommonVC}}$		0.117*** (0.028)	0.271*** (0.069)	0.248*** (0.069)
Observations	17,784	17,784	17,784	17,784
KP Wald F	33.75			
Adjusted R^2	0.72			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.098*** (0.018)	0.225*** (0.034)	0.207*** (0.033)
Observations		17,784	17,784	17,784
Adjusted R^2		0.59	0.53	0.52
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: The sample is restricted to startups located in the Silicon Valley area. Panel A reports IV estimates and Panel B reports reduced-form estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

across all three innovation outcomes. The estimates for the Silicon Valley subsample are also positive and significant. The baseline results therefore do not appear to be driven by the concentration of venture activity in Silicon Valley.

D Alternative Common VC Measure Based on Mahalanobis Technological Similarity

The baseline *CommonVC* measure uses Jaffe technological similarity to capture the technological relatedness of startups connected through common VC investors. I construct an alternative measure that replaces the Jaffe component with Mahalanobis technological similarity. Whereas Jaffe similarity depends on direct overlap across technology classes, Mahalanobis similarity also captures relationships between technologies that frequently occur together across firms.

Define the alternative technological component as

$$COMALTEC_{it} = \frac{\sum_{j \neq i} SharedVC_{ijt} \times MALTEC_{ij}^{pre}}{\sum_{j \neq i} SharedVC_{ijt}} \times 100,$$

$$MALCommonVC_{it} = \frac{COBUS_{it} + COMALTEC_{it}}{2}.$$

Thus, *MALCommonVC* differs from the baseline measure only in its technological-similarity component. Tables D1–D4 replicate the principal analyses using this alternative measure.

Taken together, Tables D1–D4 show that the main findings are not sensitive to the choice between Jaffe and Mahalanobis technological similarity in constructing *CommonVC*. The alternative measure produces a positive effect on startup innovation and technological similarity, and the geographic co-location and IP-protection interactions retain the same signs as in the baseline analysis.

Table D1: Mahalanobis Common VC Ownership and Startup Innovation

	First Stage	Second Stage		
	(1) MALCommonVC	(2) Patent Count	(3) 5-Yr Cit.	(4) 5-Yr Cit. (NS)
Treat $\times$ Post	0.759*** (0.102)			
$\widehat{\text{MALCommonVC}}$		0.091*** (0.016)	0.171*** (0.038)	0.162*** (0.033)
Observations	168,702	168,702	168,702	168,702
KP Wald F	55.46			
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: This table replicates the baseline IV analysis using *MALCommonVC*, which replaces the Jaffe-based technological component of *CommonVC* with its Mahalanobis-based alternative. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D2: Mahalanobis Common VC Ownership, Technological Similarity, and Patent Cross-Citations

	First Stage	Second Stage		
	(1) MALCommonVC	(2) Tech. Sim.	(3) Tech. Sim. (Mal.)	(4) Cross Cite
Treat $\times$ Post	0.759*** (0.102)			
$\widehat{\text{MALCommonVC}}$		0.230*** (0.048)	0.222*** (0.052)	0.036*** (0.008)
Observations	168,702	168,702	168,702	168,702
KP Wald F	55.46			
Adjusted R^2	0.66			
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: This table replicates the technological-similarity and patent-cross-citation analysis using the Mahalanobis-based common VC ownership measure. “Tech. Sim.” denotes *Technological Similarity*, and “Mal.” denotes the Mahalanobis-based alternative. *Cross Cite* measures patent cross-citations between the focal startup and startups connected through common VC investors. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D3: Mahalanobis Common VC Ownership, Geographic Co-Location, and Startup Innovation

	First Stage		Second Stage		
	(1) MALCommonVC	(2) MALCommonVC $\times$ Same CBSA	(3) Patent Count	(4) 5-Yr Cit.	(5) 5-Yr Cit. (NS)
Treat $\times$ Post	-0.027 (0.099)	-0.327*** (0.089)			
Treat $\times$ Post $\times$ Same CBSA	1.199*** (0.306)	1.522*** (0.286)			
Same CBSA	3.756*** (0.366)	4.708*** (0.358)	-0.172 (0.140)	-0.281 (0.257)	-0.234 (0.237)
$\widehat{\text{MALCommonVC}}$			0.238*** (0.061)	0.435*** (0.121)	0.425*** (0.114)
$\widehat{\text{MALCommonVC}} \times \text{Same CBSA}$			-0.153** (0.059)	-0.277*** (0.103)	-0.280*** (0.100)
Observations	168,702	168,702	168,702	168,702	168,702
KP Wald F	12.36				
Other Controls	Yes	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table replicates the geographic co-location analysis using the Mahalanobis-based common VC ownership measure. “Same CBSA” denotes *VC–Peer Same CBSA*. Columns (1) and (2) report the first stages for *MALCommonVC* and its interaction with co-location. Columns (3)–(5) report the corresponding second-stage estimates. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D4: Mahalanobis Common VC Ownership, IP Protection, and Startup Innovation

	First Stage		Second Stage		
	(1) MALCommonVC	(2) MALCommonVC $\times$ Strong IP	(3) Patent Count	(4) 5-Yr Cit.	(5) 5-Yr Cit. (NS)
Treat $\times$ Post	1.927*** (0.318)	−0.284* (0.161)			
Treat $\times$ Post $\times$ Strong IP	−1.326*** (0.325)	1.370*** (0.273)			
$\widehat{\text{MALCommonVC}}$			0.043*** (0.015)	0.117** (0.045)	0.113** (0.043)
$\widehat{\text{MALCommonVC}} \times \text{Strong IP}$			0.086*** (0.021)	0.116** (0.046)	0.108** (0.047)
Observations	110,027	110,027	110,027	110,027	110,027
KP Wald F	20.82				
Other Controls	Yes	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table replicates the IP-protection analysis using the Mahalanobis-based common VC ownership measure. “Strong IP” denotes *Strong IP Protection*. Columns (1) and (2) report the first stages for *MALCommonVC* and its interaction with Strong IP Protection. Columns (3)–(5) report the corresponding second-stage estimates. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

E No-Plausible-Rival Placebo

The identifying argument implies that COW adoption should have little effect on common VC ownership when a startup has very limited scope for economically meaningful connections to potential rivals. I examine this implication using a placebo group defined entirely from pre-adoption information.

Specifically, I classify startups as having no plausible rival when their predetermined joint technological and product market similarity to potential rivals falls in the bottom quartile among startups with positive technological similarity. This definition identifies startups for which the scope for economically meaningful common-VC connections is particularly limited, rather than startups for which rival startups are literally absent.

Table E1 compares this group with startups that have a plausible rival. Panel A shows that COW adoption has no statistically significant effect on *CommonVC* among startups in the no-plausible-rival group. The first-stage coefficient is 0.069, with an excluded-instrument F -statistic of 0.37. The reduced-form estimates for patent counts and both citation outcomes are also statistically insignificant.

Panel B shows a different pattern among startups with a plausible rival. COW adoption increases *CommonVC* by 0.771, with an excluded-instrument F -statistic of 43.79, and the reduced-form estimates are positive and statistically significant across all three innovation outcomes. Panel C estimates the two groups jointly and shows that the innovation response to COW adoption is significantly larger among startups with a plausible rival. The placebo results are therefore consistent with COW adoption affecting innovation in settings where it meaningfully changes startups' exposure to common VC ownership.

Table E1: No-Plausible-Rival Placebo

	First Stage	Reduced Form		
	(1)	(2)	(3)	(4)
	CommonVC	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Panel A: No Plausible Rival				
Treat × Post	0.069 (0.114)	0.032 (0.025)	0.057 (0.067)	0.062 (0.064)
First-Stage F (Excluded Instrument)	0.37			
Mean Dep. Var.	0.902	0.232	0.400	0.366
Observations	15,126	15,126	15,126	15,126
Adjusted R^2	0.527	0.411	0.384	0.365
Clusters	49	49	49	49
Panel B: Plausible Rival				
Treat × Post	0.771*** (0.117)	0.148*** (0.015)	0.269*** (0.027)	0.249*** (0.026)
First-Stage F (Excluded Instrument)	43.79			
Mean Dep. Var.	3.135	0.406	0.777	0.721
Observations	48,314	48,314	48,314	48,314
Adjusted R^2	0.612	0.513	0.433	0.417
Clusters	46	46	46	46
Panel C: Pooled Interaction				
Treat × Post		0.033* (0.017)	0.091** (0.036)	0.093*** (0.034)
Treat × Post × Has Rival		0.101*** (0.016)	0.146*** (0.043)	0.126*** (0.042)
Observations		65,069	65,069	65,069
Adjusted R^2		0.500	0.431	0.414
Clusters		49	49	49
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State × Year FE	Yes	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes	Yes

Notes: A startup is classified as having no plausible rival when, using pre-adoption information, its joint technological and product market similarity to potential rivals falls in the bottom quartile among startups with positive technological similarity. Panel A reports estimates for startups with no plausible rival, and Panel B reports estimates for startups with a plausible rival. Column (1) reports the first-stage effect of COW adoption on *CommonVC*; Columns (2)–(4) report reduced-form effects on innovation. Panel C pools the two groups and interacts *Treat × Post* with *Has Rival*; the main effect of *Has Rival* is absorbed by startup fixed effects. Innovation outcomes are measured as $\ln(1 + y)$. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

F Composition of the Market-Position Cells

The market position analysis divides the sample into four predetermined cells based on technological similarity and product market similarity. Table F1 reports the composition and treatment support of each cell, while Figure F1 shows their distribution in the sample.

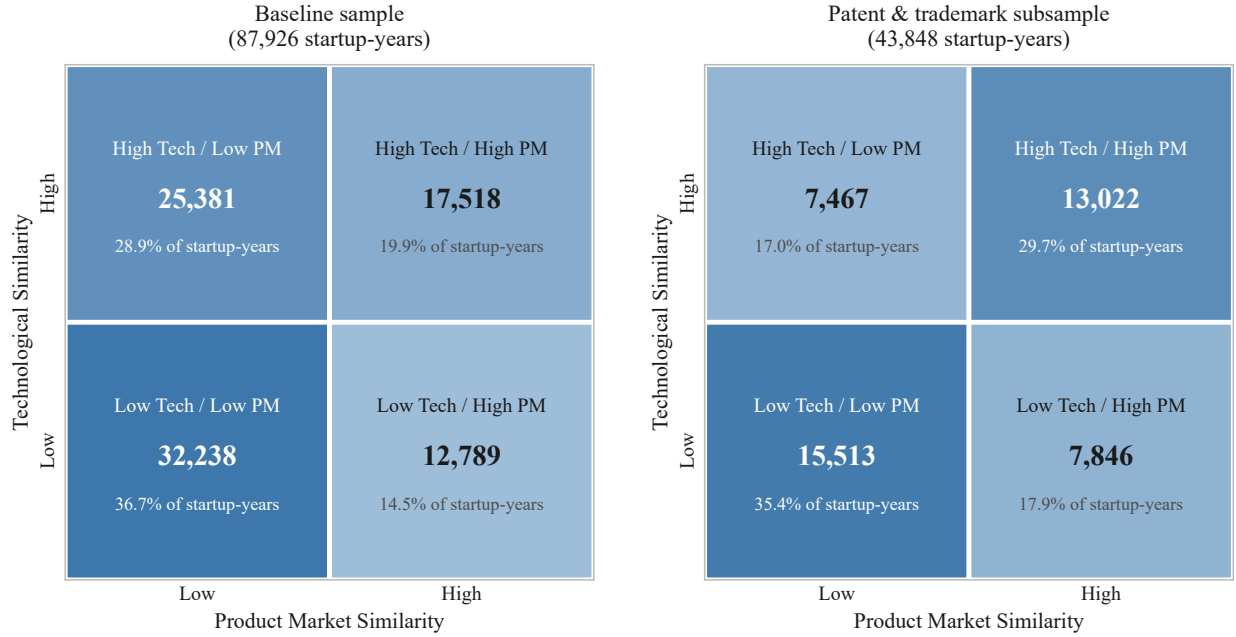
Table F1: Market-Position Cells: Support and Composition

	High Tech Low PM	High Tech High PM	Low Tech Low PM	Low Tech High PM
Panel A: Sample and Treatment Support				
Startups	1,885	1,170	2,197	861
Startup-Years	25,381	17,518	32,238	12,789
Treated Share of Startup-Years	0.251	0.180	0.431	0.276
Treated Incorporation States	8	6	5	6
Delaware Share	0.420	0.337	0.593	0.474
Panel B: Firm Characteristics				
Biopharma Share	0.313	0.217	0.225	0.074
Age	8.97	9.71	9.48	10.29
Financing Rounds per Year	0.276	0.285	0.301	0.274
Deal Value	0.484	0.592	0.537	0.571
# VC Investors	1.27	1.23	1.34	0.98
Patent Count	2.54	3.23	0.38	0.33
5-Yr Cit.	22.33	23.70	2.35	1.50
Trademark Stock	0.51	10.78	1.68	13.74
Nice Classes	0.22	2.65	0.55	3.48
Share with No Trademark	0.893	0.230	0.794	0.165
Panel C: Underlying Continuous Measures				
Baseline Tech. Sim. (Mean)	144.95	161.28	1.86	3.42
Baseline PM Sim. (Mean)	0.000	0.198	0.000	0.131

Notes: This table reports the composition of the four predetermined market position cells. “Tech” refers to predetermined *Technological Similarity*, and “PM” refers to predetermined *Product Market Similarity*. Both are measured in the last panel year at or before the startup’s COW adoption cohort. Panel A reports sample size and treatment support. Panel B reports startup-year means of firm characteristics. Panel C reports the underlying continuous measures.

Table F1 shows that the four cells are similar in startup age, financing activity, and the number of VC investors, but differ along the dimensions used to define market position. Startups in the high-tech cells patent more and receive more subsequent citations, while startups in the high-PM cells hold more trademarks and Nice classes. Each cell also contains treated observations from at least five adopting states.

Figure F1: Distribution of Startup-Years Across Market-Position Cells



Notes: Each panel reports the share of startup-years in the four predetermined market-position cells. The right panel restricts the sample to startups with at least one patent and at least one registered trademark. The median splits are recomputed within the restricted sample.

Panel C shows that the four cells differ substantially in their underlying technological and product market positions. Baseline technological similarity is much higher in the two high-tech cells than in the two low-tech cells, while baseline product market similarity is substantially higher in the two high-PM cells. Because many startups in the Low PM cells have no active trademark, I examine trademark coverage directly in the next section.

G Trademark Coverage and Product Market Similarity

The measure of *Product Market Similarity* relies on active trademarks. Startup-years with no active trademark have no observed product-market overlap under this measure and are assigned a value of zero. I therefore examine the characteristics of startups without trademarks and whether the market-position results are robust to restricting the analysis to startups with observable trademark activity.

Table G1 compares startups with and without observable trademark activity. Startups without

Table G1: Startup Characteristics by Trademark Activity

	No Trademark	Trademark Holder
Startups	10,008	7,181
Mean Entry Year	2007.5	2003.6
Years in Panel	7.73	10.62
Biopharma Share	0.295	0.202
Total Patents	9.21	15.48
Total 5-Yr Cit.	77.98	118.75
Total Financing Rounds	3.73	4.94
Max # VC Investors	2.82	3.58
Mean CommonVC	2.64	2.98
Share Entering 2015 or Later	0.253	0.136

Notes: This table compares startups that never hold an active trademark during the sample period with startups that hold at least one active trademark. Statistics are reported at the startup level.

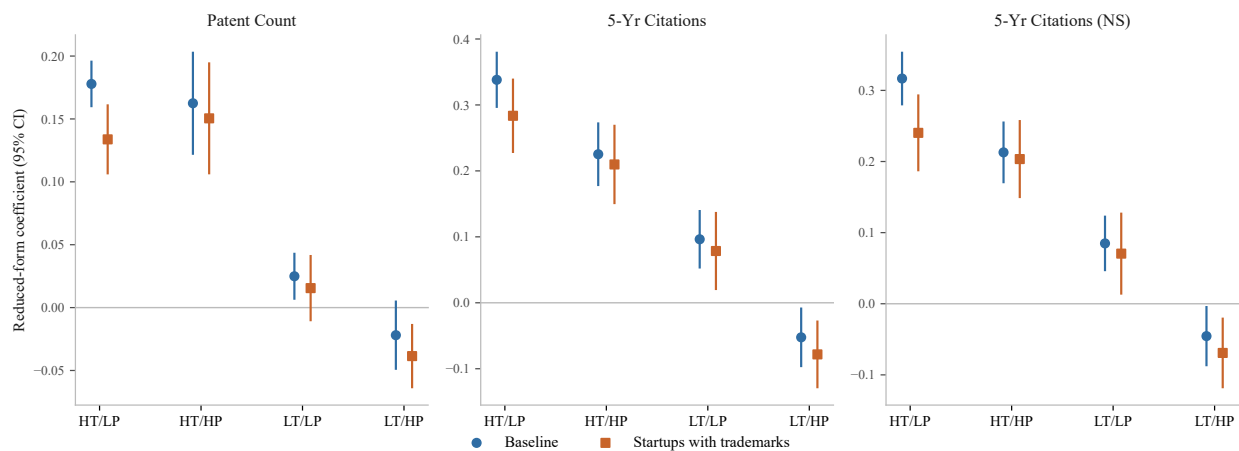
trademarks enter the panel later and have shorter observed histories. They also accumulate fewer patents and citations, complete fewer financing rounds, have fewer VC investors, and exhibit somewhat lower common VC ownership. Overall, these differences are consistent with startups without trademarks being observed at an earlier stage of development.

I restrict the sample to startups with observable trademark activity and reconstruct the market-position classification within this subsample. Table G2 shows that the market-position results remain similar in the restricted sample. The estimated responses are positive for the two high-tech cells, smaller for the Low Tech/Low PM cell, and negative and statistically significant across all three innovation outcomes for the Low Tech/High PM cell. Figure G1 shows a similar pattern in the baseline and restricted samples.

Table G2: COW Adoption and Innovation by Market Position: Startups with Trademark Activity

	(1) Patent Count	(2) 5-Yr Cit.	(3) 5-Yr Cit. (NS)
Treat $\times$ Post $\times$ High Tech / Low PM	0.134*** (0.014)	0.284*** (0.029)	0.240*** (0.028)
Treat $\times$ Post $\times$ High Tech / High PM	0.150*** (0.023)	0.210*** (0.031)	0.203*** (0.028)
Treat $\times$ Post $\times$ Low Tech / Low PM	0.015 (0.013)	0.079** (0.030)	0.070** (0.029)
Treat $\times$ Post $\times$ Low Tech / High PM	−0.039*** (0.013)	−0.078*** (0.026)	−0.069*** (0.025)
Observations	49,668	49,668	49,668
Adjusted R^2	0.471	0.410	0.396
Other Controls	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes

Notes: The sample is restricted to startups with observable trademark activity. The market-position classification is reconstructed within this subsample. The technological-similarity median is 13.46, and the product-market-similarity median is 0.048. Each reported coefficient corresponds to the interaction between *Treat* $\times$ *Post* and the indicated market-position cell. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure G1: Market-Position Reduced-Form Estimates: Baseline and Trademark Samples

Notes: This figure reports the market-position reduced-form estimates for the baseline estimation sample and for startups with observable trademark activity. For the trademark sample, the market-position classification is reconstructed within the restricted sample. Bars report 95% confidence intervals.

H Startups with Patent and Trademark Activity

I restrict the sample to startups with at least one patent and at least one registered trademark.

Table H1: Common VC Ownership and Startup Innovation: Startups with Patents and Trademarks

	First Stage	Second Stage		
	(1) CommonVC	(2) Patent Count	(3) 5-Yr Cit.	(4) 5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	0.767*** (0.142)			
$\widehat{\text{CommonVC}}$		0.065*** (0.024)	0.123** (0.050)	0.116** (0.046)
Observations	70,770	70,770	70,770	70,770
KP Wald F	29.37			
Adjusted R^2	0.63			
Panel B: Reduced-Form Estimates				
Treat $\times$ Post		0.050*** (0.018)	0.095*** (0.035)	0.089*** (0.033)
Observations		70,770	70,770	70,770
Adjusted R^2		0.45	0.39	0.37
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: The sample is restricted to startups with at least one patent and at least one registered trademark. Panel A reports the IV estimates. Column (1) reports the first-stage estimate, and Columns (2)–(4) report the corresponding second-stage estimates. Panel B reports the reduced-form estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table H1 shows that the baseline findings remain in this restricted sample. COW adoption increases *CommonVC*, with a Kleibergen–Paap Wald F -statistic of 29.37, and both the second-stage and reduced-form estimates are positive and statistically significant across all three innovation outcomes.

I A Binary Measure of Common VC Ownership

Prior work often measures common VC ownership using an indicator for whether a startup shares at least one VC investor with another startup in the same industry. I construct this indicator and compare it with the continuous *CommonVC* measure used in the main analysis. Industry is defined using PitchBook’s primary industry code. For startups covered only by VentureXpert, I manually map VentureXpert’s primary technology application class to the PitchBook classification.

The binary measure has a mean of 0.34 and a correlation of 0.49 with *CommonVC*. Among startup-years with the indicator equal to one, *CommonVC* ranges from 0.70 at the 10th percentile to 15.2 at the 90th percentile. The binary measure therefore removes substantial variation in the economic relevance of shared-VC connections.

Table II shows that the main conclusions do not depend on the continuous measure. COW adoption increases the probability that a startup shares a VC investor with a same-industry startup by 4.8 percentage points, and both the OLS and IV estimates are positive and statistically significant across all three innovation outcomes.

The two measures capture different margins. The binary measure captures whether a within-industry shared-VC connection exists, whereas *CommonVC* captures the average economic relevance of the startups connected to the focal startup through common VC investors. Because *CommonVC* averages across these connections, it abstracts from the number of shared-VC connections.

J COW Adoption and VC Characteristics

A potential concern is that COW adoption changes the characteristics or investment behavior of exposed VC investors, rather than operating only through changes in common VC ownership at the startup level. I examine this possibility using an investor-level event study around Delaware’s 2000 COW adoption.

I focus on Delaware because COW exposure is less straightforward to define at the investor level than at the startup level. A startup’s exposure is determined by its state of incorporation,

Table I1: Common VC Ownership and Startup Innovation: Binary Same-Industry Measure

	First Stage	Second Stage		
	(1)	(2)	(3)	(4)
	Binary CommonVC	Patent Count	5-Yr Cit.	5-Yr Cit. (NS)
Panel A: IV Estimates				
Treat $\times$ Post	0.048*** (0.005)			
Binary $\widehat{\text{CommonVC}}$		1.435*** (0.271)	2.691*** (0.552)	2.549*** (0.494)
Observations	168,702	168,702	168,702	168,702
KP Wald F	99.90			
Adjusted R^2	0.69			
Panel B: OLS Estimates				
Binary CommonVC		0.048*** (0.008)	0.128*** (0.009)	0.112*** (0.009)
Observations		168,702	168,702	168,702
Adjusted R^2		0.45	0.39	0.38
Other Controls	Yes	Yes	Yes	Yes
Startup FE	Yes	Yes	Yes	Yes
HQ State $\times$ Year FE	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes

Notes: *Binary CommonVC* equals one if the startup shares at least one VC investor with another startup in the same industry in that year. Industry is defined using PitchBook’s primary industry code. Startups covered only by VentureXpert are classified using their VentureXpert primary technology application class, manually mapped to the PitchBook classification. Panel A reports IV estimates. Column (1) reports the first stage, and Columns (2)–(4) report second-stage estimates in which *Binary CommonVC* is instrumented by *Treat $\times$ Post*. Panel B reports the corresponding OLS estimates. Innovation outcomes are measured as $\ln(1 + y)$. “NS” denotes citations excluding self-citations. All specifications include the full control set and startup, headquarters-state-by-year, and industry-by-year fixed effects. Standard errors are clustered by state of incorporation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

whereas a VC investor may simultaneously hold startups incorporated in multiple states that adopt COWs at different times. Constructing investor-level exposure using contemporaneous portfolio composition could therefore make exposure depend on changes in the investor's portfolio itself. Focusing on a single adoption event allows me to define investor exposure using portfolio composition fixed before the legal change.

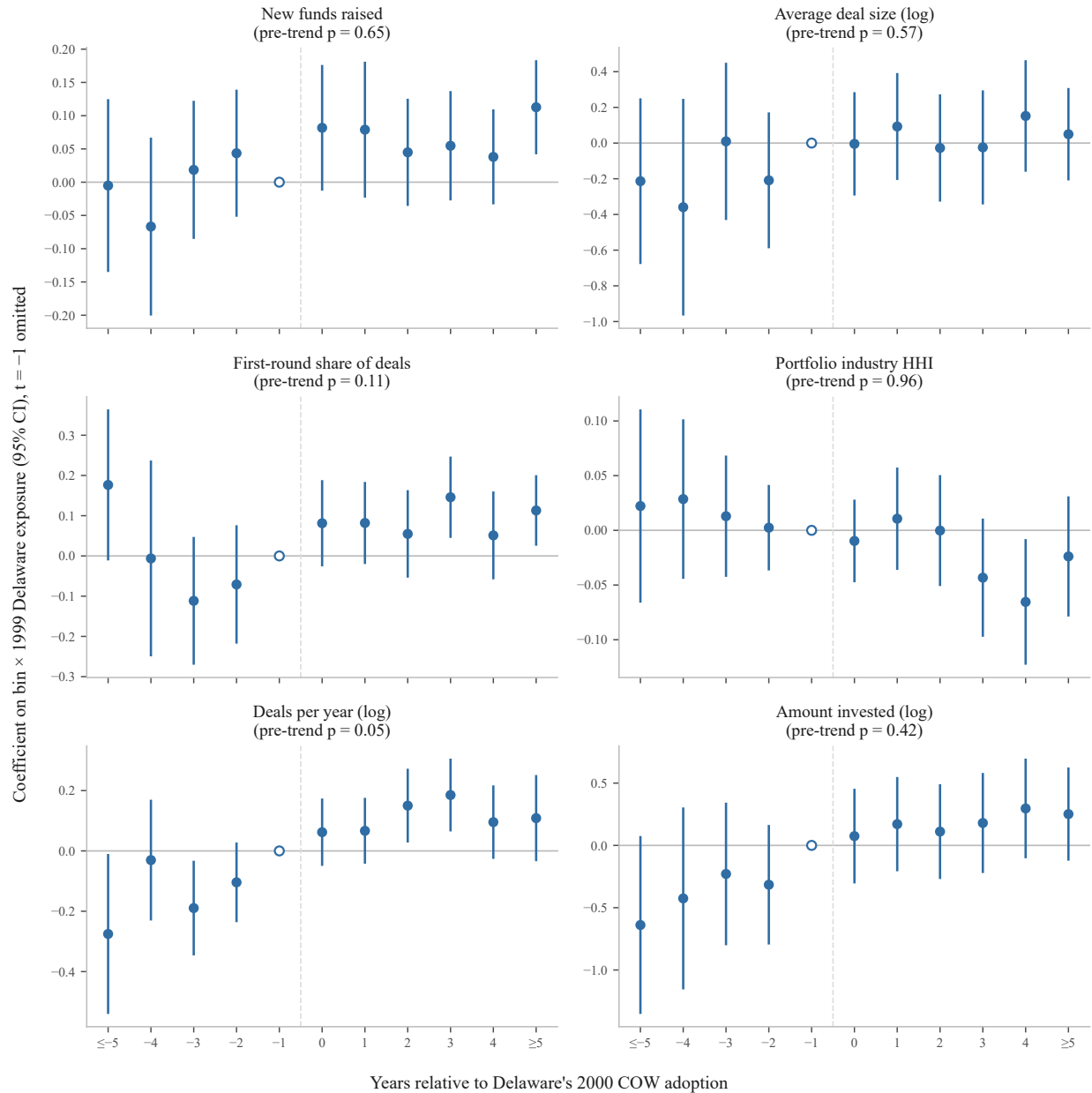
For each investor with portfolio holdings in 1999, I calculate the share of those holdings incorporated in Delaware. This continuous measure captures the investor's predetermined exposure to Delaware's 2000 COW adoption: investors with a larger share of Delaware-incorporated portfolio companies were more exposed to the legal change. Because exposure is constructed using portfolio composition in 1999, it is predetermined relative to Delaware's 2000 adoption.

Event time is defined relative to Delaware's 2000 adoption, with 1999 omitted and endpoint bins at -5 and $+5$. I interact each event-time indicator with the investor's predetermined Delaware exposure. All specifications include investor and year fixed effects, with standard errors clustered by investor. The outcomes capture fundraising (new funds raised), investment style (average deal size, the share of first-round deals, and portfolio industry concentration), and investment activity (the number of deals and total amount invested).

Figure J1 shows little evidence of broad, systematic changes in investor characteristics around Delaware's COW adoption. Most estimates are small and statistically indistinguishable from zero, although there are isolated post-adoption movements in deal counts and portfolio concentration. The first-round share and the number of new funds raised also increase modestly at longer horizons.

Overall, these results reduce the concern that the startup-level innovation effects are driven by broad contemporaneous changes in VC fundraising, investment scale, or investment style.

Figure J1: VC Characteristics Around Delaware's 2000 COW Adoption



Notes: This figure reports investor-level event-study estimates around Delaware's 2000 COW adoption. Investor exposure is predetermined using portfolio composition in 1999 and is measured as the share of the investor's 1999 portfolio holdings incorporated in Delaware. The reported coefficients correspond to event-time indicators interacted with this continuous exposure measure. The year 1999 is omitted as the reference period, and endpoint bins combine event times at or below -5 and at or above $+5$. All specifications include investor and year fixed effects. Standard errors are clustered by investor.